Trigonometric Ratios
&
Identities

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Trigonometric Ratios & Identities

Tri	Gon	Metron
↓	↓	↓
3	Sides	Measure

Basic Identities

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \operatorname{cosec}^2\theta$

$(\sec\theta - \tan\theta)$ and $(\sec\theta + \tan\theta)$ are reciprocal of each other

Proof:

We know

$1 + \tan^2\theta = \sec^2\theta$

$\Rightarrow 1 = \sec^2\theta - \tan^2\theta$

$\Rightarrow 1 = (\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$

$$\Rightarrow \sec\theta - \tan\theta = \frac{1}{\sec\theta + \tan\theta}$$

$(\operatorname{cosec}\theta - \cot\theta)$ and $(\operatorname{cosec}\theta + \cot\theta)$ are reciprocal of each other

Proof:

We know

$1 + \cot^2\theta = \operatorname{cosec}^2\theta$

$\Rightarrow 1 = \operatorname{cosec}^2\theta - \cot^2\theta$

$\Rightarrow 1 = (\operatorname{cosec}\theta - \cot\theta)(\operatorname{cosec}\theta + \cot\theta)$

$$\Rightarrow \operatorname{cosec}\theta - \cot\theta = \frac{1}{(\operatorname{cosec}\theta + \cot\theta)}$$ Hence Proved.

Note: The values of following trigonometric ratios must be remembered.

	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0

	0°	30°	45°	60°	90°
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	ND
cot	ND	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	ND
cosec	ND	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

Q1 **Find the value of $\frac{\tan^2 A \sin^2 A}{\tan^2 A - \sin^2 A}$ if $0° < A < 90°$**

Sol.

$$\tan^2 A \cdot \sin^2 A = \tan^2 A (1 - \cos^2 A)$$
$$= \tan^2 A - \tan^2 A \cdot \cos^2 A$$
$$= \tan^2 A - \frac{\sin^2 A}{\cos^2 A} \cdot \cos^2 A$$
$$= \tan^2 A - \sin^2 A$$

Hence $\frac{\tan^2 A \sin^2 A}{\tan^2 A - \sin^2 A} = 1$

Q2 **Find the value of $\frac{\cot^2 A \cos^2 A}{\cot^2 A - \cos^2 A}$ if $0° < A < 90°$**

Sol.

$$\cot^2 A \cdot \cos^2 A = \cot^2 A (1 - \sin^2 A)$$
$$= \cot^2 A - \cot^2 A \cdot \sin^2 A$$
$$= \cot^2 A - \frac{\cos^2 A}{\sin^2 A} \cdot \sin^2 A$$
$$= \cot^2 A - \cos^2 A$$

Hence $\frac{\cot^2 A \cos^2 A}{\cot^2 A - \cos^2 A} = 1$

Q3 **Prove that** $\left(\frac{1+\sin\alpha}{1+\cos\alpha}\right)\left(\frac{1+\sec\alpha}{1+\operatorname{cosec}\alpha}\right) = \tan\alpha$

Sol. From LHS $\left(\frac{1+\sin\alpha}{1+\cos\alpha}\right)\left(\frac{1+1/\cos\alpha}{1+1/\sin\alpha}\right)$

$\Rightarrow \left(\frac{1+\sin\alpha}{1+\cos\alpha}\right)\left(\frac{\cos\alpha+1}{\cos\alpha}\right)\left(\frac{\sin\alpha}{\sin\alpha+1}\right)$

$\Rightarrow \frac{\sin\alpha}{\cos\alpha} = \tan\alpha$ = RHS **Hence proved.**

Q4 **Prove that** $\frac{\sin x + \cos x}{\cos^3 x} = \tan^3 x + \tan^2 x + \tan x + 1$

Sol. In LHS multiply by $\sin^2 x + \cos^2 x$

LHS $\Rightarrow \frac{(\sin x + \cos x)(\sin^2 x + \cos^2 x)}{\cos^3 x}$ $(\because \sin^2 x + \cos^2 x = 1)$

$\Rightarrow \frac{\sin^3 x}{\cos^3 x} + \frac{\sin x \cdot \cos^2 x}{\cos^3 x} + \frac{\cos x \cdot \sin^2 x}{\cos^3 x} + \frac{\cos^3 x}{\cos^3 x}$

$\Rightarrow \tan^3 x + \tan x + \tan^2 x + 1$ = RHS **Hence proved.**

Q5 **Prove that** $\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta,\ 0 < \theta < \frac{\pi}{2}$

Sol. LHS $\Rightarrow \sqrt{\frac{1+\sin\theta}{1-\sin\theta} \times \frac{1+\sin\theta}{1+\sin\theta}} = \sqrt{\frac{(1+\sin\theta)^2}{1-\sin^2\theta}}$

$\Rightarrow \sqrt{\frac{(1+\sin\theta)^2}{\cos^2\theta}} = \frac{1+\sin\theta}{\cos\theta} = \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}$

$= \sec\theta + \tan\theta$ = RHS **Hence proved.**

Q6 **If $\tan\theta + \sec\theta = 1.5$ then find $\sin\theta$, $\tan\theta$, $\sec\theta$.**

Sol. $\sec\theta + \tan\theta = \frac{3}{2}$ ...(1)

We know that $\sec\theta - \tan\theta = \frac{1}{\sec\theta + \tan\theta}$

Hence $\sec\theta - \tan\theta = \frac{2}{3}$ ...(2)

(1) + (2) $\quad 2\sec\theta = \frac{13}{6} \Rightarrow \sec\theta = \frac{13}{12}$

(1) − (2) $\quad 2\tan\theta = \frac{5}{6} \Rightarrow \tan\theta = \frac{5}{12}$

$\sin\theta = \frac{\tan\theta}{\sec\theta} = \frac{5}{13}$

Q7 **Prove that $\frac{1}{\text{secA} - \text{tanA}} - \frac{1}{\text{cosA}} = \frac{1}{\text{cosA}} - \frac{1}{\text{secA} + \text{tanA}}$**

Sol. $\because \frac{1}{\text{secA} - \text{tanA}} = \text{secA} + \text{tanA}$

$\therefore$ LHS is (secA + tanA) − (secA)

= tanA ...(1)

Now $\because \frac{1}{\text{secA} + \text{tanA}} = \text{secA} - \text{tanA}$

$\therefore$ RHS is (secA) − (secA − tanA)

= tanA = LHS **Hence proved.**

Q8 **Prove that $(\sec\theta + \text{cosec}\theta)(\sin\theta + \cos\theta) = \sec\theta\,\text{cosec}\theta + 2$**

Sol. $\text{LHS} = \left\{\frac{1}{\cos\theta} + \frac{1}{\sin\theta}\right\}(\sin\theta + \cos\theta)$

$\Rightarrow \frac{(\sin\theta + \cos\theta)^2}{\sin\theta\cos\theta} \Rightarrow \frac{\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta}{\sin\theta\cos\theta}$

$\Rightarrow \frac{1 + 2\sin\theta\cos\theta}{\sin\theta\cos\theta} \Rightarrow \frac{1}{\sin\theta\cos\theta} + 2$

$\Rightarrow \sec\theta\,\text{cosec}\theta + 2 = \text{RHS}$ **Hence proved.**

Q9 **Prove that: $(\sec^2\theta + \tan^2\theta)(\text{cosec}^2\theta + \cot^2\theta) = 1 + 2\sec^2\theta\,\text{cosec}^2\theta$**

Sol.

$$\text{LHS} \Rightarrow \left\{\frac{1}{\cos^2\theta} + \frac{\sin^2\theta}{\cos^2\theta}\right\}\left\{\frac{1}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta}\right\}$$

$$\Rightarrow \left\{\frac{1+\sin^2\theta}{\cos^2\theta}\right\}\left\{\frac{1+\cos^2\theta}{\sin^2\theta}\right\}$$

$$\Rightarrow \left\{\frac{2-\cos^2\theta}{\cos^2\theta}\right\}\left\{\frac{2-\sin^2\theta}{\sin^2\theta}\right\}$$

$$\Rightarrow \frac{4 - 2\left(\sin^2\theta + \cos^2\theta\right) + \sin^2\theta\cos^2\theta}{\sin^2\theta \cdot \cos^2\theta}$$

$$\Rightarrow \frac{2 + \sin^2\theta\cos^2\theta}{\sin^2\theta \cdot \cos^2\theta}$$

$\Rightarrow 2\sec^2\theta.\,\text{cosec}^2\theta + 1 = \text{RHS}$ **Hence proved.**

Q10 **If $\sin\theta + \sin^2\theta = 1$, then find the value of $\cos^{12}\theta + 3\cos^{10}\theta + 3\cos^8\theta + \cos^6\theta - 1$**

Sol. $\because \sin\theta = 1 - \sin^2\theta$

$\sin\theta = \cos^2\theta$ ...(i)

Now given expression can be written as

$\cos^6\theta\,\{\cos^6\theta + 3\cos^4\theta + 3\cos^2\theta + 1\} - 1$

$\Rightarrow \sin^3\theta\,\{\sin^3\theta + 3\sin^2\theta + 3\sin\theta + 1\} - 1$ [From equation (i)]

$\Rightarrow \sin^3\theta\,(\sin\theta + 1)^3 - 1$

$\Rightarrow \{\sin^3\theta\,(\sin\theta + 1)\}^3 - 1$

$\Rightarrow (\sin\theta + \sin^2\theta)^3 - 1$

$\Rightarrow 1 - 1 = \mathbf{0}$

Q11 **Show that $2(\sin^6x + \cos^6x) - 3(\sin^4x + \cos^4x) + 1 = 0$**

Sol. $\sin^6x + \cos^6x = (\sin^2x)^3 + (\cos^2x)^3$

$= (\sin^2x + \cos^2x)^3 - 3\sin^2x\cos^2x\,(\sin^2x + \cos^2x)$

$= 1 - 3\sin^2x\cos^2x$ .....(i)

Now

$\sin^4x + \cos^4x = (\sin^2x)^2 + (\cos^2x)^2$

$= (\sin^2x + \cos^2x)^2 - 2\sin^2x\cos^2x$

$= 1 - 2\sin^2x\cos^2x$ .....(ii)

Now LHS of given expression

$2(1 - 3\sin^2x\cos^2x) - 3(1 - 2\sin^2x\cos^2x) + 1$

$\Rightarrow 2 - 6\sin^2x\cos^2x - 3 + 6\sin^2x\cos^2x + 1 = 0 = \text{RHS}$

Q12 **If $3\sin\theta + 5\cos\theta = 5$ then show that $5\sin\theta - 3\cos\theta = \pm 3$**

Sol. $\because\ 3\sin\theta + 5\cos\theta = 5$

on squaring

$\Rightarrow 9\sin^2\theta + 25\cos^2\theta + 30\sin\theta\cos\theta = 25$

$\Rightarrow 9(1 - \cos^2\theta) + 25(1 - \sin^2\theta) + 30\sin\theta\cos\theta = 25$

$\Rightarrow 25\sin^2\theta + 9\cos^2\theta - 30\sin\theta\cos\theta = 9$

$\Rightarrow (5\sin\theta - 3\cos\theta)^2 = 3^2$

$\Rightarrow 5\sin\theta - 3\cos\theta = \pm 3$ **Hence proved.**

Q13 **The circumference of a circle circumscribing an equilateral triangle is 24π units. Find the area of the circle inscribed in the equilateral triangle.**

Sol. Let ABC is an equilateral triangle and O is center of its circumcircle,

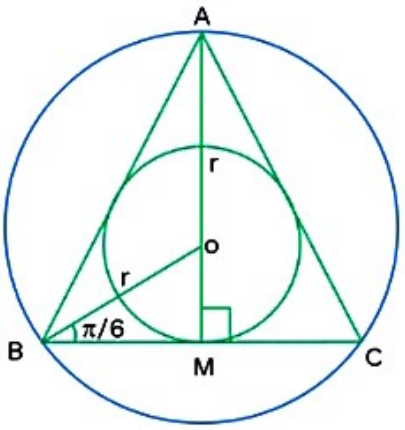
also its radius = r

$\therefore\ 2\pi r = 24\pi \Rightarrow r = 12$

In ΔOBM

$$BM = r\cos\frac{\pi}{6} = 12\frac{\sqrt{3}}{2} = 6\sqrt{3}$$

$$OM = r\sin\frac{\pi}{6} = 12\left(\frac{1}{2}\right) = 6$$



Incircle will be concentric with circumcircle and its radius will be OM = 6

Area of incircle = $\pi(6)^2 = 36\pi$ unit

Q14 **Two circles of radii 4cm and 1cm touch each other externally and θ is angle contained by their direct common tangents. Find $\sin\theta$**

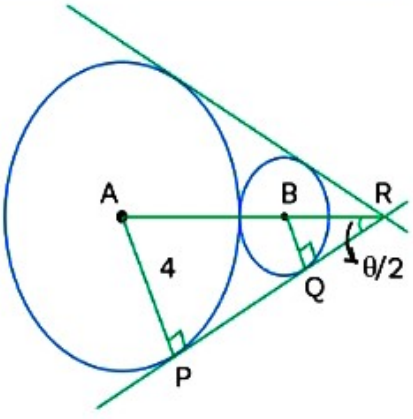
Sol. In ΔAPR

$$AP = AR\sin\frac{\theta}{2}$$

$$AR = \frac{AP}{\sin\frac{\theta}{2}} \Rightarrow AR = \frac{4}{\sin\frac{\theta}{2}} \quad \text{.....(i)}$$

in ΔBQR

$$BQ = BR\sin\frac{\theta}{2} \Rightarrow BR = \frac{1}{\sin\frac{\theta}{2}} \quad \text{.....(ii)}$$



(i) – (ii)

$$(AR - BR) = \frac{3}{\sin\frac{\theta}{2}} \Rightarrow AB = \frac{3}{\sin\frac{\theta}{2}}$$

$$\sin\frac{\theta}{2} = \frac{3}{AB} = \frac{3}{4+1} = \frac{3}{5}$$

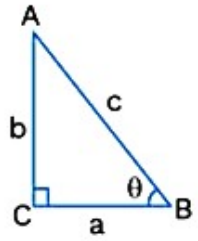
Hence $\cos\frac{\theta}{2} = \frac{4}{5}$; $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25}$

Q15 **If in ΔABC, $\angle C = 90°$ then prove that $\tan A + \tan B = \frac{c^2}{ab}$ (Where a, b, c are sides opposite to angles A, B, C respectively)**

Sol. In ΔABC

$$\tan A = \frac{BC}{AC} = \frac{a}{b} \quad \text{.....(i)}$$

$$\tan B = \frac{AC}{BC} = \frac{b}{a} \quad \text{.....(ii)}$$



(i) + (ii)

$$\tan A + \tan B = \frac{a}{b} + \frac{b}{a} \Rightarrow \frac{a^2 + b^2}{ab}$$

$$= \frac{c^2}{ab} \quad \{\because a^2 + b^2 = c^2\}$$

Basics of trigonometry and Measurement of angles

Measure of Angles:

Measure of an angle means the amount of turning. There are two types of measures to measure an angle, which are commonly used.

(i) Sexagesimal measure

(ii) Radian measure

Sexagesimal Measurement (The British System)

In this system

180 degree or 180° = 2 right angle

Circular (radian) Measurement

In this system

π radian or π^c = 2 right angle

$\Rightarrow$ 1 radian = $\frac{2}{\pi}$ right angle = constant

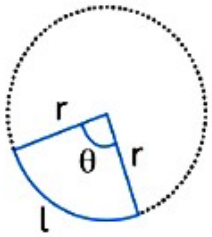
1 radian = $\frac{180^\circ}{\pi} = 57^\circ$ $\quad (\pi = 3.141592)$

1 degree = $\frac{\pi}{180}$ radian

Q **Express 150° in terms of radian.**

Sol. $\because\ 1^\circ = \frac{\pi}{180}$radian $\quad \therefore\ 150^\circ = \frac{\pi}{180} \times 150$radian $= \frac{5\pi}{6}$radian

Note:



(i) A radian is constant angle

(ii) 1 right angle = $\frac{\pi}{2}$radian = 90°

(iii) 2 right angles = π radians = 180°

(iv) $\frac{\pi}{2} = 90^\circ, \frac{\pi}{3} = 60^\circ, \frac{\pi}{6} = 30^\circ, \frac{\pi}{4} = 45^\circ, \frac{3\pi}{2} = 270^\circ, 2\pi = 360^\circ$

(v) If no unit of measurement is indicated for an angle, it will be understood that radian measure is implied.

Note:

If θ be the angle subtended by an arc of length 'l' on the centre of a circle of radius 'r' then

→ $l = r\theta$

→ Area of the sector = $\frac{1}{2}lr = \frac{1}{2}r^2\theta$

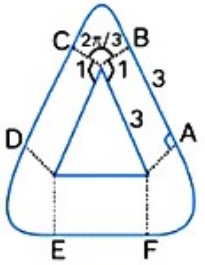
Q1 **Consider an equilateral triangle with sides 3 cm. Now if a man runs around the triangle in such a way that he is always at a distance of 1 cm from the sides of triangle then how much distance will he travel.**

Sol. According to given figure total distance travelled by man

$$= AB + CD + EF + \overset{\frown}{BC} + \overset{\frown}{DE} + \overset{\frown}{FA}$$

$$= 3+3+3+\left(1\times\frac{2\pi}{3}\right)+\left(1\times\frac{2\pi}{3}\right)+\left(1\times\frac{2\pi}{3}\right)$$

$$= (9+2\pi) \text{ unit}$$



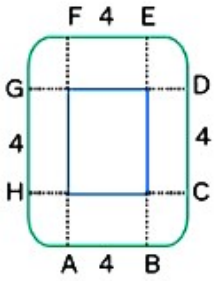
Q2 **Consider a square of sides 4 cm. Now if a man runs at a distance of 1 cm from the side of square. How much distance will he travel?**

Sol. According to given figure total distance travelled by man

$$= \left(AB + CD + EF + GH\right) + \left(\overset{\frown}{BC} + \overset{\frown}{DE} + \overset{\frown}{FG} + \overset{\frown}{HA}\right)$$

$$= 4(4) + 4\left(\frac{\pi}{2}\times 1\right)$$

$$\Rightarrow (16+2\pi)\text{unit}$$



Q3 **Consider a triangle with sides 3, 6, 8 cm respectively. Now if a man runs around the triangle in such a way that he is always at a distance of 1 cm from the sides of triangle then how much distance will he travel?**

Sol. Total travelled distance

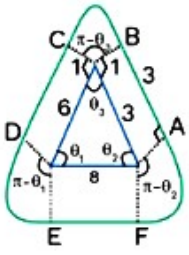
$$= AB + CD + EF + \left(\overset{\frown}{BC} + \overset{\frown}{DE} + \overset{\frown}{FA}\right)$$

$$= 3 + 6 + 8 + 1\times(\pi-\theta_3) + 1\times(\pi-\theta_2) + 1\times(\pi-\theta_1)$$

$$= 17 + 3\pi - (\theta_1+\theta_2+\theta_3)$$

$$= 17 + 3\pi - (\pi)$$

$$\Rightarrow (17+2\pi)\text{unit}$$



Q4 **A pod rack has 3 straight sides of length 12 cm each, rounded of by three circular arcs each of 3 cm radius as shown in the figure. If the area inside the rack can be expressed as $a\pi + b\sqrt{3} + c$ where a, b, c are whole numbers then the value of (a + b + c) is**

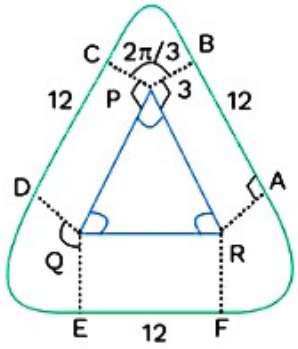
Sol. Area inside = ar(ΔPQR) + 3ar(PBAR) + 3 area of sector PBC

$$= \frac{\sqrt{3}}{4}(12)^2 + 3\times(12\times 3) + 3\left(3^2 \times \frac{2\pi}{3}\times\frac{1}{2}\right)$$

$$= 36\sqrt{3} + 9\pi + 108$$

$\therefore$ a = 9, b = 36, c = 108

a + b + c = **153**



Real definition of 2 basic functions (sine & cosine) Trigonometric ratios of any magnitude:

A point P lies on circle such that it makes an angle θ with the x-axis. It is at a distance of x from y-axis and at a distance of y from x-axis and r is the radius of the circle then

$$\sin\theta = \frac{\text{Distance of point P from } x-\text{axis with direction}}{\text{Radius of the circle}}$$

$$\cos\theta = \frac{\text{Distance of point P from } y-\text{axis with direction}}{\text{Radius of the circle}}$$

Limiting Cases

(i) $\theta \to 0,\quad y \to 0,\quad x \to r$

$\therefore \sin 0 = \frac{0}{r} = 0,\ \cos 0 = \frac{r}{r} = 1$

(ii) $\theta \to 90°,\quad x \to 0,\quad y \to r$

$\therefore \sin 90° = \frac{r}{r} = 1,\ \cos 90° = \frac{0}{r} = 0$

(iii) $\theta = 30°$

$$\sin 30° = \frac{1}{2},\ \cos 30° = \frac{\sqrt{3}}{2}$$

(iv) $\theta = 45°$ then

$$\sin 45° = \frac{1}{\sqrt{2}},\ \cos 45° = \frac{1}{\sqrt{2}}$$

(v) $\theta = 60°$ then

$$\sin 60° = \frac{\sqrt{3}}{2},\ \cos 60° = \frac{1}{2}$$

Reduction Formulae

(I) "90° – θ"

$\sin(90° - \theta) = \cos\theta$	$\cos(90° - \theta) = \sin\theta$
$\tan(90° - \theta) = \cot\theta$	$\cot(90° - \theta) = \tan\theta$
$\sec(90° - \theta) = \operatorname{cosec}\theta$	$\operatorname{cosec}(90° - \theta) = \sec\theta$

Q1 **Show that:**

$\sin^2 5° + \sin^2 10° + \sin^2 15° + \ldots\ldots + \sin^2 90° = 9\frac{1}{2}$

Sol. LHS

$$\left(\sin^2 5^\circ + \sin^2 85^\circ\right) + \left(\sin^2 10^\circ + \sin^2 80^\circ\right) + \cdots + \left(\sin^2 40^\circ + \sin^2 50^\circ\right) + \sin^2 45^\circ + \sin^2 90^\circ$$

$$\Rightarrow \left(\sin^2 5^\circ + \cos^2 5^\circ\right) + \left(\sin^2 10^\circ + \cos^2 10^\circ\right) + \cdots + \left(\sin^2 40^\circ + \cos^2 40^\circ\right) + \sin^2 45^\circ + \sin^2 90^\circ$$

$$1 + 1 + 1 + \ldots (9 \text{ times}) + \left(\frac{1}{\sqrt{2}}\right)^2$$

$$\Rightarrow 9\frac{1}{2} = \text{RHS}$$

Hence proved.

Q2 **Show that: $\tan1°.\tan2°.\tan3°\ldots\ldots\tan89° = 1$**

Sol. LHS $\left(\tan1^\circ \times \tan89^\circ\right) \times \left(\tan2^\circ \times \tan88^\circ\right) \times \ldots\ldots \times \left(\tan44^\circ \times \tan46^\circ\right) \times \left(\tan45^\circ\right)$

$$\Rightarrow \left(\tan1^\circ \times \cot1^\circ\right) \times \left(\tan2^\circ \times \cot2^\circ\right) \times \ldots\ldots \times \left(\tan44^\circ \times \cot44^\circ\right) \times 1$$

$\Rightarrow 1 \times 1 \times 1\ldots\ldots = 1$ RHS

Hence proved.

Q3 **Find the value of $\cos^2\frac{\pi}{16} + \cos^2\frac{3\pi}{16} + \cos^2\frac{5\pi}{16} + \cos^2\frac{7\pi}{16}$**

Sol. $\cos\frac{7\pi}{16} = \sin\left(\frac{\pi}{2} - \frac{7\pi}{16}\right) = \sin\frac{\pi}{16}$

$\cos\frac{5\pi}{16} = \sin\left(\frac{\pi}{2} - \frac{5\pi}{16}\right) = \sin\frac{3\pi}{16}$

Hence give exp.

$\cos^2\frac{\pi}{16} + \cos^2\frac{3\pi}{16} + \sin^2\frac{3\pi}{16} + \sin^2\frac{\pi}{16}$

$\left(\cos^2\frac{\pi}{16} + \sin^2\frac{\pi}{16}\right) + \left(\cos^2\frac{3\pi}{16} + \sin^2\frac{3\pi}{16}\right)$

$= 1 + 1 = \mathbf{2}$

Q4 **Show that:** $\cot\frac{\pi}{20}\cot\frac{3\pi}{20}\cot\frac{5\pi}{20}\cot\frac{7\pi}{20}\cot\frac{9\pi}{20}=1$

Sol. $\cot\frac{9\pi}{20}=\tan\left(\frac{\pi}{2}-\frac{9\pi}{20}\right)=\tan\frac{\pi}{20}$

$\cot\frac{7\pi}{20}=\tan\left(\frac{\pi}{2}-\frac{7\pi}{20}\right)=\tan\frac{3\pi}{20}$

Hence LHS

$\cot\frac{\pi}{20}\cdot\cot\frac{3\pi}{20}\cdot\cot\frac{\pi}{4}\cdot\tan\frac{3\pi}{20}\cdot\tan\frac{\pi}{20}$

$\left(\cot\frac{\pi}{20}\cdot\tan\frac{\pi}{20}\right)\cdot\left(\cot\frac{3\pi}{20}\cdot\tan\frac{3\pi}{20}\right)\cdot\cot\frac{\pi}{4}$

$= 1 \times 1 \times 1 =$ RHS **Hence proved.**

(II) "90° + θ"

$\sin(90^\circ+\theta)=\cos\theta$ $\quad$ $\cos(90^\circ+\theta)=-\sin\theta$

$\tan(90^\circ+\theta)=-\cot\theta$ $\quad$ $\cot(90^\circ+\theta)=-\tan\theta$

$\sec(90^\circ+\theta)=-\operatorname{cosec}\theta$ $\quad$ $\operatorname{cosec}(90^\circ+\theta)=\sec\theta$

(III) "180° − θ"

$\sin(180^\circ-\theta)=\sin\theta$ $\quad$ $\cos(180^\circ-\theta)=-\cos\theta$

$\tan(180^\circ-\theta)=-\tan\theta$ $\quad$ $\cot(180^\circ-\theta)=-\cot\theta$

$\sec(180^\circ-\theta)=-\sec\theta$ $\quad$ $\operatorname{cosec}(180^\circ-\theta)=\operatorname{cosec}\theta$

Q1 **Evaluate: (i)** $\sin(120°)$ **(ii)** $\cos(150°)$

Sol. **(i)** $\sin(120°)$

$= \sin\left(180^0 - 60^0\right)$

$= \sin 60^0 = \frac{\sqrt{3}}{2}$

(ii) $\cos(150°)$

$= \cos\left(180 - 30^\circ\right)$

$= -\cos 30^\circ = -\frac{\sqrt{3}}{2}$

Q2 **Prove that:**
$\cos 10° + \cos 20° + \cos 30° + \cos 80° + \cos 100° + \cos 150° + \cos 160° + \cos 170° = 0$

Sol. $\cos(100°) = \cos(180° - 80°) = -\cos 80°$

$\cos(150°) = \cos(180° - 30°) = -\cos 30°$

$\cos(160°) = \cos(180° - 20°) = -\cos 20°$

$\cos(170°) = \cos(180° - 10°) = -\cos 10°$

$\therefore$ required exp. = 0

Q3 **Find the value of** $\tan\frac{\pi}{11} + \tan\frac{2\pi}{11} + \tan\frac{4\pi}{11} + \tan\frac{7\pi}{11} + \tan\frac{9\pi}{11} + \tan\frac{10\pi}{11}$.

Sol. $\tan\frac{7\pi}{11} = \tan\left(\pi - \frac{4\pi}{11}\right) = -\tan\frac{4\pi}{11}$

$\tan\frac{9\pi}{11} = \tan\left(\pi - \frac{2\pi}{11}\right) = -\tan\frac{2\pi}{11}$

$\tan\frac{10\pi}{11} = \tan\left(\pi - \frac{\pi}{11}\right) = -\tan\frac{\pi}{11}$

$\therefore$ required exp. = **0**

Q4 **If A, B, C, D are angles of cyclic quadrilateral then prove that**
cosA + cosB + cosC + cosD = 0

Sol. Let A, B, C, D are in order then A + C =π, B + D = π
cosA = cos(π − C) = − cosC
cosB = cos(π − D) = − cosD
cosA + cosB + cosC + cosD = 0

(IV) "180° + θ"

sin(180° + θ) = − sinθ
tan(180° + θ) = tanθ
sec(180° + θ) = − secθ

cos(180° + θ) = − cosθ
cot(180° + θ) = cotθ
cosec(180° + θ) = − cosecθ

Q **Evaluate:**

(i) **tan(225°)**
= tan(180° + 45°)
= tan45°
= 1

(ii) **sin(210°)**
= sin(180° + 30°)
= − sin30°
$= -\frac{1}{2}$

(iii) **cos(240°)**
= cos(180° + 60°)
= −cos60°
$= -\frac{1}{2}$

(iv) **sin(270°)**
= sin(180° + 90°)
= −sin90°
= − 1

(v) **cos(270°)**
= cos(180° + 90°)
= −cos90°
= 0

(vi) **True / False sin4 is negative**
$\because\ \pi < 4 < \frac{3\pi}{2} \Rightarrow 4 \in$ **III**
Quadrant
$\therefore$ **sin4 < 0 ⇒ True**

(V) "$-\theta$" or "$(360° - \theta)$"

$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$
$\tan(-\theta) = -\tan\theta$	$\cot(-\theta) = -\cot\theta$
$\sec(-\theta) = \sec\theta$	$\operatorname{cosec}(-\theta) = -\operatorname{cosec}\theta$

Q **Evaluate:**

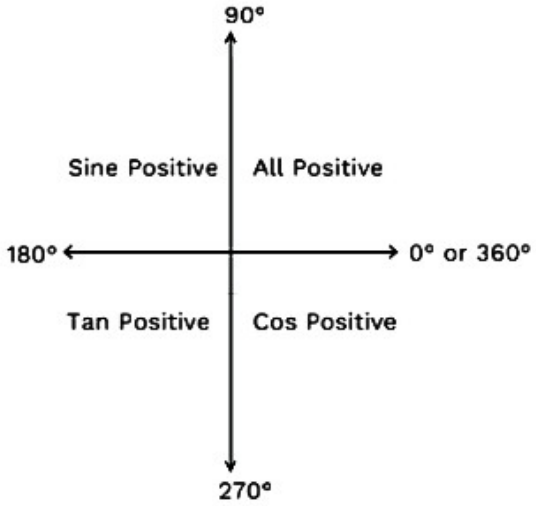
(i) $\tan(-30°)$
$= -\tan30°$
$= -\frac{1}{\sqrt{3}}$

(ii) $\sec(-60°)$
$= \sec(60°)$
$= 2$

(iii) $\cos(315°)$
$= \cos(360° - 45°)$
$= \cos45° = \frac{1}{\sqrt{2}}$

(iv) $\tan(330°)$
$= \tan(360° - 30°)$
$= -\tan30° = -\frac{1}{\sqrt{3}}$

To remember the sign rule use



Q1 **Evaluate:**

(i) $\tan(-120°)$
$= -\tan(180° - 60°)$
$= -(-\tan60°) = \sqrt{3}$

(ii) $\cos(180°)$
$= \cos(90° + 90°)$
$= -\sin90°$
$= -1$

Q2 **Prove that: $\sin(-420°)\cos(390°) + \cos(-660°)\sin(330°) = -1$**

Sol.

$$\sin(-420°) = -\sin(420°) = -\sin(360° + 60°)$$
$$= -\sin 60° = -\frac{\sqrt{3}}{2}$$

$$\cos(390°) = \cos(360° + 30°)$$
$$= \cos 30° = \frac{\sqrt{3}}{2}$$

$$\cos(-660°) = \cos(660°)$$
$$= \cos(720° - 60°)$$
$$= \cos 60°$$
$$= \frac{1}{2}$$

$$\sin(330°) = \sin(360° - 30°)$$
$$= -\sin 30°$$
$$= -\frac{1}{2}$$

$$\therefore \text{LHS} = \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)$$

$$= -\frac{3}{4} - \frac{1}{4} = -1 = \text{RHS}$$

Hence proved.

Q3 **If $\sin^2\theta_1 + \sin^2\theta_2 + \sin^2\theta_3 = 0$ then which of the following is not the possible value of $\cos\theta_1 + \cos\theta_2 + \cos\theta_3$**

(A) 3 **(B) –3** **(C) –1** **(D) –2**

Sol. **(D)**

$$\sin^2\theta_1 + \sin^2\theta_2 + \sin^2\theta_3 = 0$$

hence from given equation it is clear that
$\sin^2\theta_1 = \sin^2\theta_2 = \sin^2\theta_3 = 0$

$\theta_1, \theta_2, \theta_3 \in n\pi, n \in I$

Now $\cos\theta_1, \cos\theta_2, \cos\theta_3 \in \{-1, 1\}$

hence $\cos\theta_1 + \cos\theta_2 + \cos\theta_3$ can be **{–3, –1, 1, 3}**

Q4 **Which of the following is possible?**

(A) $\sin\theta = \frac{5}{3}$ **(B)** $\tan\theta = 1002$

(C) $\cos\theta = \frac{1+p^2}{1-p^2}\left(p \neq \pm 1\right)$ **(D)** $\sec\theta = \frac{1}{2}$

Sol. **(BC)**

$\because -1 \le \sin\theta \le 1 \Rightarrow (A)\text{false}$

$\because \tan\theta \in R \quad \Rightarrow (B)\text{true}$

$\because -1 \le \cos\theta \le 1,$ for $p = 0,\ \frac{1+p^2}{1-p^2} = 1 \Rightarrow (C)$ true

$\because \sec\theta \in (-\infty, -1] \cup [1, \infty) \Rightarrow (D)\text{false}$

Q5 **If x = sin1, y = sin1° then show that x > y.**

Sol. $1 = \frac{180°}{\pi}$ hence

$1 > 1° \Rightarrow$ **sin1 > sin1°**

($\because$ 1, 1° both lies in I quadrant and sine increase from 0 to $\pi/2$)

Q6 **Which of the following is greatest?**

(A) tan1 **(B) tan4** **(C) tan7** **(D) tan10**

Sol. **(A)**

$\because \tan\theta = \tan(\theta - n\pi), n \in I$

$\therefore \tan 4 = \tan(4 - \pi)$

$\therefore \tan 7 = \tan(7 - 2\pi)$

$\therefore \tan 10 = \tan(10 - 3\pi)$

now $0 < 10 - 3\pi < 7 - 2\pi < 4 - \pi < 1$

$\tan 0 < \tan(10 - 3\pi) < \tan(7 - 2\pi) < \tan(4 - \pi) < \tan 1$

$0 < \tan 10 < \tan 7 < \tan 4 < \tan 1$

7 **Is the equation $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$ possible for real values of x and y?**

Sol. We know that $\sec^2\theta \geq 1$

$\therefore \dfrac{4xy}{(x+y)^2} \geq 1$

$\Rightarrow 4xy \geq (x+y)^2$

$\Rightarrow (x+y)^2 - 4xy \leq 0$

$\Rightarrow (x-y)^2 \leq 0 \Rightarrow (x-y) = 0$

$\Rightarrow x = y$

$\therefore x = y \in R - \{0\}$

So, equation is possible for **x = y**.

Q8 **Show that the equation $\sin\theta = x + \dfrac{1}{x}$ is impossible if x is real.**

Sol. From given equation $x(\sin\theta) = x^2 + 1$

$\Rightarrow x^2 - (\sin\theta)x + 1 = 0$

Which is quadratic equation in x also $x \in R$

$\therefore D \geq 0$

$(-\sin\theta)^2 - 4(1)(1) \geq 0$

$\sin^2\theta - 4 \geq 0 \Rightarrow \sin^2\theta \geq 4$

Which is not possible.

Q9 **If $\operatorname{cosec}\theta = \dfrac{x^2 - y^2}{x^2 + y^2}$ where x, y are two unequal non-zero real numbers then prove that θ has no real value.**

Sol. $\operatorname{cosec}\theta \leq -1$ or $\operatorname{cosec}\theta \geq 1$

$\dfrac{x^2 - y^2}{x^2 + y^2} \leq -1$ or $\dfrac{x^2 - y^2}{x^2 + y^2} \geq 1$

$x^2 - y^2 \leq -x^2 - y^2$ or $x^2 - y^2 \geq x^2 + y^2$

$2x^2 \le 0$ or $2y^2 \le 0$

$x^2 = 0$ $y^2 = 0$

$x = 0$ $y = 0$

(not possible) (not possible)

$\therefore$ θ has no real values.

Important table

Radian	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
Degree	0°	30°	45°	60°	90°	120°	135°	150°	180°
sin	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0	−1/2	$-1/\sqrt{2}$	$-\sqrt{3}/2$	−1
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	N.D.	$-\sqrt{3}$	−1	$-1/\sqrt{3}$	0
cot	N.D	$\sqrt{3}$	1	$1/\sqrt{3}$	0	$-1/\sqrt{3}$	−1	$-\sqrt{3}$	N.D

$7\pi/6$	$5\pi/4$	$4\pi/3$	$3\pi/2$	$5\pi/3$	$7\pi/4$	$11\pi/6$	2π
210°	225°	240°	270°	300°	315°	330°	360°
−1/2	$-1/\sqrt{2}$	$-\sqrt{3}/2$	−1	$-\sqrt{3}/2$	$-1/\sqrt{2}$	−1/2	0
$-\sqrt{3}/2$	$-1/\sqrt{2}$	−1/2	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1
$1/\sqrt{3}$	1	$\sqrt{3}$	N.D.	$-\sqrt{3}$	−1	$-1/\sqrt{3}$	0
$\sqrt{3}$	1	$1/\sqrt{3}$	0	$-1/\sqrt{3}$	−1	$-\sqrt{3}$	N.D

Highlight of the table

(i) $-1 \le \sin\theta \le 1$

(ii) $-1 \le \cos\theta \le 1$

$\tan\theta$, $\cot\theta$ attain all values from $(-\infty, \infty)$ i.e. any real value.

(iii) $\sin(n\pi) = 0$ where $n \in I$

$\therefore$ Sine of integral multiple of π is zero.

(iv) $\tan(n\pi) = 0$ where $n \in I$

$\therefore$ tangent of integral multiple of π is zero.

(v) $\cos(2n-1)\frac{\pi}{2} = 0$ where $n \in I$

$\therefore$ cosine of odd integral multiple of $\frac{\pi}{2}$ is zero.

(vi) $\cot(2n-1)\frac{\pi}{2} = 0$ where $n \in I$

$\therefore$ cotangent of odd integral multiple of $\frac{\pi}{2}$ is zero.

(vii) $\cos 2m\pi = 1$; $\cos(2m-1)\pi = -1$ where $m \in I$
cosine of odd integral multiple of π is -1.
$\sec 2m\pi = 1$; $\sec(2m-1)\pi = -1$

(viii) $\tan(2n+1)\frac{\pi}{2}$ = N.D. where $n \in I$

$\therefore$ tangent of odd integral multiple of $\frac{\pi}{2}$ is N.D.

(ix) $\cot(n\pi)$ = N.D. where $n \in I$
$\therefore$ cotangent of integral multiple of π is N.D.

(x) $\sin\left(2n\pi + \frac{\pi}{2}\right) = 1, \sin\left(2n\pi - \frac{\pi}{2}\right) = -1 = \sin\left(2n\pi + \frac{3\pi}{2}\right)$

(xi) cos is decreasing function in 1st quadrant

(xii) For $\theta \in \left(0, \frac{\pi}{4}\right)$

$\cos\theta > \sin\theta$, $\cot\theta > \tan\theta$, $\operatorname{cosec}\theta > \sec\theta$

Q1 **Which one is incorrect:**

(A) $\sqrt{2.56} < 0.125 \times 13.6$

(B) $\sqrt{0.9} < 0.9$

(C) $\frac{1}{\sec^2 10} + \frac{1}{\operatorname{cosec}^2 10} = 1$

(D) (26 × 28) + 1 is perfect square of a natural number.

Sol. **(B)**

(A) $\sqrt{2.56} = \sqrt{\frac{256}{100}} = \frac{16}{10} = 1.6$

$0.125 \times 13.6 = \frac{125}{1000} \times \frac{136}{10} = \frac{136}{8 \times 10} = \frac{17}{10} = 1.7$ (A) is true

(B) for $0 < x < 1 \Rightarrow \sqrt{x} > x$ (B) → False

(C) $\cos^2 10 + \sin^2 10 = 1$ (C) → True

(D) $(27-1)(27+1) + 1 \Rightarrow 27^2 - 1 + 1 = 27^2$ (D) → True

Q2 **Find the distance between orthocenter and circumcenter in a triangle with sides 17, 15, 8.**

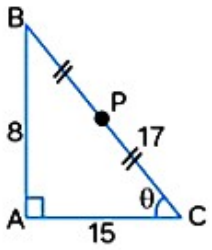
Sol. Clearly $17^2 = 289$

$15^2 + 8^2 = 255 + 64 = 289$

$\therefore \Delta$ is right angled orthocenter will be A

Circumcenter will be mid point of BC which is P

Now $AP = \frac{1}{2}BC = \frac{17}{2}$ **unit**



Q3 **Find the cosine of interior angle of a regular dodecagon.**

Sol. One interior angle of n-sided regular polygon is $\frac{(n-2)\pi}{n}$

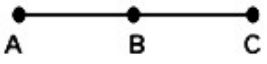
$\therefore$ for dodecagon $\theta = \frac{(12-2)}{12}\pi = \frac{5\pi}{6} = 150°$

$\therefore \cos\theta = \cos 150° = -\frac{\sqrt{3}}{2}$

Q4 **How many circles of finite radius can be drawn through 3-collinear points.**

Sol. If A, B, C are points which lies on any circle then AB and BC will be chord of that circle and perpendicular bisector of AB and BC will be diameters which will never intersect as they will be parallel.

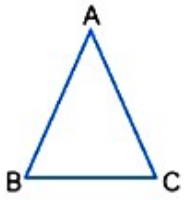
$\therefore$ **No such circle possible**.



Q5 **How many circles of finite radius can be drawn through 3-non collinear points.**

Sol. If a circle passes through points A, B, C then it will circumcircle of ΔABC which can be only one.

$\therefore$ **One such circle is possible**.

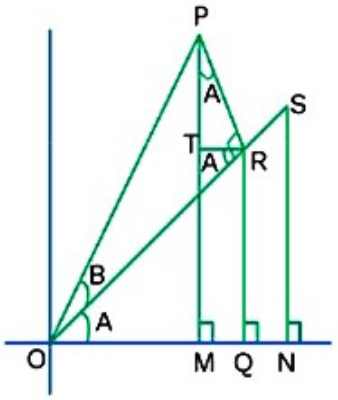


Trigonometric ratios of compound angles

Note:

Trigonometric ratios i.e., sine, cosine, tan, cot, sec and cosec are not distributed over addition and subtraction of 2-angles

i.e., $\sin(A+B) \neq \sin A + \sin B$



(I) sin(A + B) = sinA cosB + cosA sinB

This equation is gold of trigonometry.

Proof:

$$\sin(A+B) = \frac{PM}{OP} = \frac{PT+TM}{OP} = \frac{PT}{OP} + \frac{RQ}{OP}$$

$$= \frac{PT}{PR}\cdot\frac{PR}{OP} + \frac{RQ}{OR}\cdot\frac{OR}{OP}$$

$$= \cos A \sin B + \sin A \cos B$$

Hence proved.

Ex.: $\sin 99° \cos 21° + \cos 99° \sin 21°$

$= \sin(99° + 21°)$

$= \sin(120°) = \frac{\sqrt{3}}{2}$

$\{\because \cos(-B) = \cos B, \sin(-B) = -\sin B\}$

(II) sin(A−B) = sinA cosB − cosA sinB

Proof:

In in formula of sin (A+B) replace B by −B

$\sin(A+(-B)) = \sin A \cos(-B) + \cos A \sin(-B)$

$\sin(A-B) = \sin A \cos B - \cos A \sin B$

Hence proved.

(III) cos(A + B) = cosA cosB − sinA sinB

Proof:

In formula of $\sin(A + B) = \sin A \cos B + \cos A \sin B$

substitute A by $\frac{\pi}{2} + A$ $\quad \sin\left(\frac{\pi}{2} + A + B\right) = \sin\left(\frac{\pi}{2} + A\right)\cos B + \cos\left(\frac{\pi}{2} + A\right)\sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

Hence proved.

(IV) cos(A−B) = cosA cosB + sinA sinB

Proof:

In formula of cos(A + B)

substitute B by − B

$\cos(A-B) = \cos A \cos(-B) - \sin A \sin(-B)$

$\cos(A-B) = \cos A \cos B + \sin A \sin B$

Hence proved.

Note:

By geometrical interpretation we can prove that:

(i) $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

(ii) $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$\because$ AB = CD

AB = CP + PD

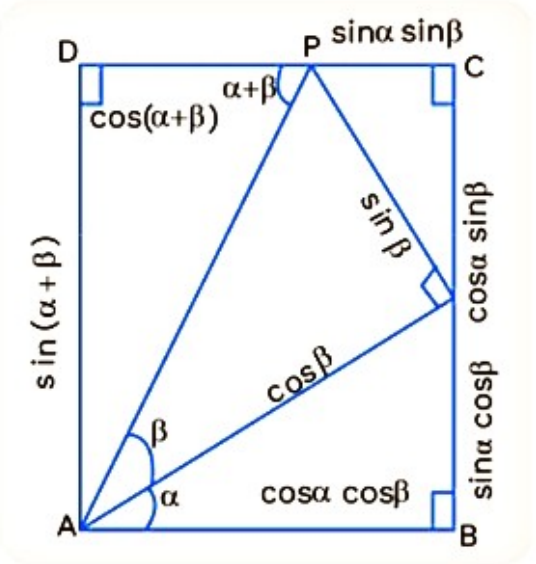
$\cos\alpha\cos\beta = \sin\alpha\sin\beta + \cos(\alpha+\beta)$

$\Rightarrow \cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

AD = BC

AD = BQ + QC

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$



To deduce the value of trigonometric ratios of 75° and 15°

$\sin(15°) = \sin(45° - 30°)$

$= \sin45°\cos30° - \cos45°\sin30°$

$= \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\cdot\frac{1}{2} = \frac{\sqrt{3}-1}{2\sqrt{2}}$

$\sin(75°) = \sin(45° + 30°)$

$= \sin45°\cos30° + \cos45°\sin30°$

$= \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}\cdot\frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}}$

Note:

1. $\sin\frac{\pi}{12} = \sin15° = \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4} = \cos75° = \cos\frac{5\pi}{12}$

2. $\sin\frac{5\pi}{12} = \sin75° = \frac{\sqrt{3}+1}{2\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{4} = \cos15° = \cos\frac{\pi}{12}$

3. $\tan15° = \tan\frac{\pi}{12} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3} = \cot75° = \cot\frac{5\pi}{12}$

4. $\tan75° = \tan\frac{5\pi}{12} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3} = \cot15° = \cot\frac{\pi}{12}$

Q1 **Find the value of sin105°**

Sol. $\sin 105^\circ = \sin(60^\circ + 45^\circ)$

$= \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ$

$= \frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{2}} + \frac{1}{2}\cdot\frac{1}{\sqrt{2}}$

$= \frac{\sqrt{3}+1}{2\sqrt{2}}$

Q2 **Find the value of cos165°**

Sol. $\cos(165^\circ) = \cos(120^\circ + 45^\circ)$

$= \cos 120^\circ \cos 45^\circ - \sin 120^\circ \sin 45^\circ$

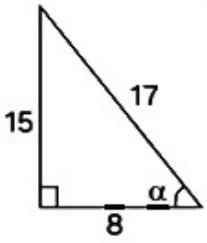
$= \left(-\frac{1}{2}\right)\cdot\frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{2}}$

$= \frac{-(\sqrt{3}+1)}{2\sqrt{2}}$

Q3 **Given $\sin\alpha = \frac{15}{17}, \cos\beta = \frac{-5}{13}$, find $\sin(\alpha - \beta)$**

Sol. $\therefore \sin\alpha = \frac{15}{17}, \quad \Rightarrow \cos\alpha = \pm\frac{8}{17}$

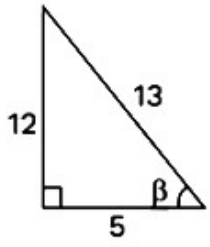
$\therefore \cos\beta = \frac{-5}{13} \quad \Rightarrow \sin\beta = \pm\frac{12}{13}$



$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

$= \frac{15}{17}\left(\frac{-5}{13}\right) - \left(\pm\frac{8}{17}\right)\left(\pm\frac{12}{13}\right)$

$\frac{-75}{221} \pm \frac{96}{221} = \frac{21}{221}$ **or** $\frac{-171}{221}$



Q4 **If $\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-\alpha)=-\frac{3}{2}$ then prove that**

$\cos\alpha+\cos\beta+\cos\gamma=\sin\alpha+\sin\beta+\sin\gamma=0$

Sol. From given equation

$2\cos(\alpha-\beta)+2\cos(\beta-\gamma)+2\cos(\gamma-\alpha)+1+1+1=0$

$(\sin\alpha^2+\cos^2\alpha)+(\sin^2\beta+\cos^2\beta)+(\sin^2\gamma+\cos^2\gamma)+(2\cos\alpha\cos\beta+2\sin\alpha\sin\beta)$

$+(2\cos\beta\cos\gamma+2\sin\beta\sin\gamma)+(2\cos\gamma\cos\alpha+2\sin\gamma\sin\alpha)=0$

$\Rightarrow(\sin\alpha+\sin\beta+\sin\gamma)^2+(\cos\alpha+\cos\beta+\cos\gamma)^2=0$

$\Rightarrow\sin\alpha+\sin\beta+\sin\gamma=\cos\alpha+\cos\beta+\cos\gamma=0$

Two very important identities

(a) $\sin(A+B)\cdot\sin(A-B)=\sin^2A-\sin^2B$
$=\cos^2B-\cos^2A$

(b) $\cos(A+B)\cdot\cos(A-B)=\cos^2A-\sin^2B$
$=\cos^2B-\sin^2A$

Next four identities (Converting product into sum)

(i) $2\sin A\cos B=\sin(A+B)+\sin(A-B)$

(ii) $2\cos A\sin B=\sin(A+B)-\sin(A-B)$

(iii) $2\cos A\cos B=\cos(A+B)+\cos(A-B)$

(iv) $2\sin A\sin B=\cos(A-B)-\cos(A+B)$

Note:

These four identities are useful for converting product into sum.

Q1 **Find the value of expression $\cos^2 73°+\cos^2 47°+\cos73°\cos47°$**

Sol. $\cos^2 73°-\sin^2 47°+1+\frac{1}{2}(2\cos73°\cos47°)$

$\cos(73°+47°)\cos(73°-47°)+1+\frac{1}{2}(\cos120°+\cos26°)$

$\cos(120°)\cos26°+1+\frac{1}{2}\left(-\frac{1}{2}+\cos26°\right)$

$\left(\frac{-1}{2}\right)\cos26°+1-\frac{1}{4}+\frac{1}{2}\cos26°=1-\frac{1}{4}=\frac{3}{4}$

Identities for converting sum into product.

(i) $\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(ii) $\sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(iii) $\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(iv) $\cos C - \cos D = 2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{D-C}{2}\right)$

$= -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

Q1 **Find the value of** $\left(\frac{\sin 75^\circ - \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ}\right)^8$.

Sol. $\Rightarrow \left(\frac{2\cos 45^\circ \sin 30^\circ}{2\cos 45^\circ \cos 30^\circ}\right)^8 \Rightarrow \left(\frac{1/2}{\sqrt{3}/2}\right)^8 = \left(\frac{1}{\sqrt{3}}\right)^8 \Rightarrow \frac{1}{3^4} = \frac{1}{81}$

Q2 **Prove that $\cos 80^\circ + \cos 40^\circ - \cos 20^\circ = 0$**

Sol. LHS

$2\cos\left(\frac{80^\circ + 40^\circ}{2}\right)\cos\left(\frac{80^\circ - 40}{2}\right) - \cos 20^\circ$

$\Rightarrow 2\cos 60^\circ \cos 20^\circ - \cos 20^\circ$

$\Rightarrow 2.\left(\frac{1}{2}\right)\cos 20^\circ - \cos 20^\circ$

$\Rightarrow \cos 20^\circ - \cos 20^\circ = 0 = \text{RHS}$ **Hence proved.**

Q3 **Find the value of expression** $\frac{\sin 8\theta \cos\theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos\theta - \sin 3\theta \sin 4\theta}$ **given** $\theta = 75^\circ$

Sol. $\frac{\sin 8\theta \cos\theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos\theta - \sin 3\theta \sin 4\theta}$

$\Rightarrow \frac{(\sin 9\theta + \sin 7\theta) - (\sin 9\theta + \sin 3\theta)}{(\cos 3\theta + \cos\theta) - (\cos\theta - \cos 7\theta)}$

$$\Rightarrow \frac{\sin 7\theta - \sin 3\theta}{\cos 3\theta + \cos 7\theta} \quad \Rightarrow \frac{2\cos 5\theta \sin 2\theta}{2\cos 5\theta \cos 2\theta} \Rightarrow \tan 2\theta = \tan(150°)$$

$$= \tan(180° - 30°) = -\tan 30° = -\frac{1}{\sqrt{3}}$$

Q4 **If cos x + cos y + cosα = 0 and sin x + sin y + sinα = 0 then find the value of $\left(\frac{x+y}{2}\right)$ in terms of α**

Sol. cos x + cos y = − cosα

$$\Rightarrow 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) = -\cos\alpha \qquad ...(i)$$

sin x + sin y = − sin α

$$\Rightarrow 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) = -\sin\alpha \qquad ...(ii)$$

(i) ÷ (ii)

$$\cot\left(\frac{x+y}{2}\right) = \cot\alpha \Rightarrow \frac{x+y}{2} = \alpha$$

Q5 **Determine the smallest positive value of x (in degrees) for which tan(x + 100°) = tan(x + 50°) tanx tan(x − 50°)**

Sol. $\frac{\tan(x+100°)}{\tan x} = \tan(x+50°)\tan(x-50°)$

$$\frac{2\sin(x+100°)\cos x}{2\cos(x+100°)\sin x} = \frac{2\sin(x+50°)\sin(x-50°)}{2\cos(x+50°)\cos(x-50°)}$$

$$\frac{\sin(2x+100°)+\sin 100°}{\sin(2x+100°)-\sin 100°} = \frac{\cos 100° - \cos 2x}{\cos 100° + \cos 2x}$$

by C/D

$$\frac{\sin(2x+100°)}{\sin 100°} = -\frac{\cos 100°}{\cos 2x}$$

$\Rightarrow$ 2sin(2x + 100°)cos2x = −2sin100°cos100°

$\Rightarrow$ sin(4x + 100°) + sin100° = −sin200°

$\Rightarrow$ sin(4x + 100°) = −(sin200° + sin100°)

$= -2\sin 150° \cos 50° = -2.\frac{1}{2}\cos 50°$

$\Rightarrow \quad \sin(4x + 100°) = -\cos 50° = \sin(270° - 50°)$

$\Rightarrow \quad \sin(4x + 100°) = \sin 220°$

$\Rightarrow \quad 4x + 100° = 220°$

$\Rightarrow \quad 4x = 120°$

$\Rightarrow \quad \mathbf{x = 30°}$

Q6 **Let k = 1° then prove that**

$$\sum_{n=0}^{88} \frac{1}{\cos nk \cos(n+1)k} = \frac{\cos 1°}{\sin^2 1°}$$

Sol.

$$\frac{1}{\cos nk\, \cos(n+1)k} = \frac{1}{\sin k} \frac{\sin\{(n+1)k - nk\}}{\cos nk\, \cos(n+1)k}$$

$$= \frac{1}{\sin k} \frac{\sin(n+1)k \cos nk - \cos(n+1)k \sin nk}{\cos nk\, \cos(n+1)k}$$

$$= \frac{1}{\sin k}\{\tan(n+1)k - \tan(nk)\}$$

$$\therefore \sum_{n=0}^{88} \frac{1}{\cos nk \cos(n+1)k} = \frac{1}{\sin k}\left\{\sum_{n=0}^{88} \tan(n+1)k - \tan nk\right\}$$

$$= \frac{1}{\sin k}\{\tan 89k\} = \frac{\tan 89°}{\sin 1°} \Rightarrow \frac{\cot 1°}{\sin 1°}$$

$$= \frac{\cos 1°}{\sin^2 1°}$$

To deduce the value of tan(A + B) and cot (A + B)

1. $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
2. $\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$
3. $\tan(\pi/4 + \theta) = \dfrac{1 + \tan\theta}{1 - \tan\theta}$
4. $\tan(\pi/4 - \theta) = \dfrac{1 - \tan\theta}{1 + \tan\theta}$
5. $\cot(A+B) = \dfrac{\cot B \cot A - 1}{\cot B + \cot A}$
6. $\cot(A-B) = \dfrac{\cot B \cot A + 1}{\cot B - \cot A}$

Q1 **$x - y = \frac{\pi}{4}$ and cotx + coty = 2. Find smallest positive angles x and y**

Sol. $\cot(x-y) = \cot\left(\frac{\pi}{4}\right)$

$$\frac{\cot y \cot x + 1}{\cot y - \cot x} = 1$$

$$\Rightarrow \cot y \cot x + 1 = \cot y - \cot x$$

$$\Rightarrow (2 - \cot x)\cot x + 1 = (2 - \cot x) - (\cot x)$$

$$\Rightarrow (\cot x)^2 - 4(\cot x) + 1 = 0$$

$$\Rightarrow \cot x = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$$

$\Rightarrow$ If x, y are smallest positive angle then $\cot x = 2 - \sqrt{3} \Rightarrow x = 75°$ and $x - y = 45° \Rightarrow$ **y = 30°**

Q2 **Prove that: cot16° cot44° + cot76° (cot44° – cot16°) = 3**

Sol. LHS cot16° cot44° + cot76° (cot44° – cot16°)

$$\cot(16° + 44°) = \frac{\cot 44° \cot 16° - 1}{\cot 44° + \cot 16°}$$

$$\Rightarrow (\cot 44° + \cot 16°)\frac{1}{\sqrt{3}} + 1 = \cot 44° \cot 16° \quad \text{...(i)}$$

$$\cot(76° + 44°) = \frac{\cot 44° \cot 76° - 1}{\cot 44° + \cot 76°}$$

$$\Rightarrow (\cot 44° + \cot 76°)\left(\frac{-1}{\sqrt{3}}\right) + 1 = \cot 44° \cot 76° \quad \text{...(ii)}$$

$$\cot(76° - 16°) = \frac{\cot 16° \cot 76° + 1}{\cot 16° - \cot 76°}$$

$$\Rightarrow (\cot 16° - \cot 76°)\frac{1}{\sqrt{3}} - 1 = \cot 16° \cot 76° \quad \text{...(iii)}$$

(i) + (ii) – (iii)

cot16° cot44° + cot76° (cot44° – cot16°) = 3

= RHS **Hence proved.**

Q3 **If tanA − tanB = x and cotA − cotB = y then prove that cot (A − B) = $\frac{1}{x} - \frac{1}{y}$**

Sol. $\frac{1}{\cot A} - \frac{1}{\cot B} = x \Rightarrow \frac{\cot B - \cot A}{\cot A \cot B} = x$

$$\frac{1}{x} = \frac{(\cot A \cot B + 1) - 1}{\cot B - \cot A} = \cot(A - B) - \frac{1}{\cot B - \cot A}$$

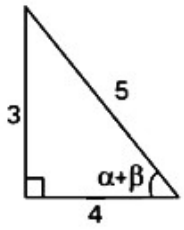
$$\frac{1}{x} = \cot(A - B) - \frac{1}{(-y)}$$

$$\cot(A - B) = \frac{1}{x} - \frac{1}{y}$$

Hence proved.

Q4 **If $\cos(\alpha + \beta) = \frac{4}{5}, \sin(\alpha - \beta) = \frac{5}{13}$ and $\alpha, \beta \in \left(0, \frac{\pi}{4}\right)$ then find the value of tan2α**

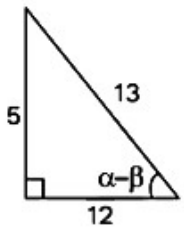
Sol. $\alpha + \beta \in \left(0, \frac{\pi}{2}\right)$ and $\alpha - \beta \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$



Now $\cos(\alpha + \beta) = \frac{4}{5}$

$\Rightarrow \tan(\alpha + \beta) = \frac{3}{4}$

also $\sin(\alpha - \beta) = \frac{5}{13}$



$\Rightarrow \tan(\alpha - \beta) = \frac{5}{12}$

Now $\tan 2\alpha = \tan\{(\alpha + \beta) + (\alpha - \beta)\}$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta)\tan(\alpha - \beta)}$$

$$= \frac{3/4 + 5/12}{1 - \frac{3}{4}\cdot\frac{5}{12}} = \frac{56}{33}$$

Q5 If $m\tan(\theta-30^\circ)=n\tan(\theta+120^\circ)$ then show that $\cos 2\theta=\dfrac{m+n}{2(m-n)}$

Sol. $\dfrac{\tan(\theta+120^\circ)}{\tan(\theta-30^\circ)}=\dfrac{m}{n}$

$$\frac{\sin(\theta+120^\circ)\cos(\theta-30^\circ)}{\cos(\theta+120^\circ)\sin(\theta-30^\circ)}=\frac{m}{n}$$

Now use C/D

$$\frac{\sin(\theta+120^\circ)\cos(\theta-30^\circ)+\cos(\theta+120^\circ)\sin(\theta-30^\circ)}{\sin(\theta+120^\circ)\cos(\theta-30^\circ)-\cos(\theta+120^\circ)\sin(\theta-30^\circ)}=\frac{m+n}{m-n}$$

$$\frac{\sin(\theta+120^\circ+\theta-30^\circ)}{\sin(\theta+120^\circ-\theta+30^\circ)}=\frac{m+n}{m-n}$$

$$\frac{\sin(2\theta+90^\circ)}{\sin(150^\circ)}=\frac{m+n}{m-n}$$

$$\frac{\cos 2\theta}{1/2}=\frac{m+n}{m-n}\Rightarrow\cos 2\theta=\frac{m+n}{2(m-n)}$$

T-RATIOS OF MULTIPLE AND SUBMULTIPLE ANGLES

Multiple angles are $2\theta,3\theta,4\theta$ etc. and sub-multiple angles are $\dfrac{\theta}{2},\dfrac{\theta}{4},\dfrac{\theta}{8}$ etc.

1. $\sin2A=2\sin A\cos A$

$\Rightarrow 1+\sin2A=(\sin A+\cos A)^2$

$\Rightarrow 1-\sin2A=(\sin A-\cos A)^2$

2. $\sin2A=\dfrac{2\tan A}{1+\tan^2A}$

3. $\cos2A=\cos^2A-\sin^2A$

$=1-2\sin^2A$

$=2\cos^2A-1$

$\Rightarrow\dfrac{1}{2}(1+\cos2A)=\cos^2A$

$\Rightarrow\dfrac{1}{2}(1-\cos2A)=\sin^2A$

4. $\cos2A=\dfrac{1-\tan^2A}{1+\tan^2A}$

$\Rightarrow\tan^2A=\dfrac{1-\cos2A}{1+\cos2A}$

5. $\tan2A=\dfrac{2\tan A}{1-\tan^2A}$

Q1 **Asking**

(i) **cosA = 1/3, find cos2A**

(ii) **If sinA = 2/5, find cos2A**

(iii) **sinA = 3/5, find sin2A**

(iv) **tanA = 1/3, find tan2A**

(v) **tanA = 1/7, find sin2A**

(vi) **$\cos A = \frac{4}{5}$, find $\tan\frac{A}{2}$**

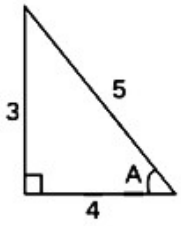
Sol.

(i) $\cos 2A = 2\cos^2 A - 1 = 2\left(\frac{1}{9}\right) - 1 = -\frac{7}{9}$

(ii) $\cos 2A = 1 - 2\sin^2 A$

$$= 1 - 2\left(\frac{4}{25}\right) = \frac{17}{25}$$

(iii) $\because \sin A = \frac{3}{5} \quad \therefore \cos A = \pm\frac{4}{5}$



$\sin 2A = 2\sin A \cos A$

$$= 2\left(\frac{3}{5}\right)\left(\pm\frac{4}{5}\right) = \pm\frac{24}{25}$$

(iv) $\because \tan 2A = \frac{2\tan A}{1 - \tan^2 A}$

$$= \frac{2(1/3)}{1 - (1/9)} = \frac{2/3}{8/9} = \frac{3}{4}$$

(v) $\because \sin 2A = \frac{2\tan A}{1 + \tan^2 A}$

$$= \frac{2(1/7)}{1 + 1/49} = \frac{2/7}{50/49} = \frac{7}{25}$$

(vi) $\cos A = \frac{1 - \tan^2 A/2}{1 + \tan^2 A/2}$

$$\Rightarrow \frac{4}{5} = \frac{1 - \tan^2 A/2}{1 + \tan^2 A/2}$$

$$\Rightarrow 4 + 4\tan^2 A/2 = 5 - 5\tan^2 A/2$$

$$\Rightarrow 9\tan^2 A/2 = 1 \Rightarrow \tan A/2 = \pm\frac{1}{3}$$

To deduce the value of tan22.5°

Let $x = 22.5°$

$\Rightarrow 2x = 45°$

$\tan 2x = \tan 45° = 1$

$$\frac{2\tan x}{1-\tan^2 x} = 1 \Rightarrow \tan^2 x + 2\tan x - 1 = 0$$

$$\Rightarrow \tan x = \frac{-2 \pm \sqrt{4+4}}{2} = -1 \pm \sqrt{2}$$

$\because x \in$ I quadrant

$\therefore \tan x = -1 + \sqrt{2} = \tan 22.5°$

Note:

(i) $\tan\frac{\pi}{8} = \sqrt{2} - 1 = \cot\frac{3\pi}{8}$

(ii) $\tan\frac{3\pi}{8} = \sqrt{2} + 1 = \cot\frac{\pi}{8}$

Q1 **Prove that** $\frac{\sec 8\theta - 1}{\sec 4\theta - 1} = \frac{\tan 8\theta}{\tan 2\theta}$

Sol. LHS $\frac{1/\cos 8\theta - 1}{1/\cos 4\theta - 1}$

$$\Rightarrow \frac{1-\cos 8\theta}{1-\cos 4\theta} \times \frac{\cos 4\theta}{\cos 8\theta}$$

$$\Rightarrow \frac{2\sin^2 4\theta}{2\sin^2 2\theta} \times \frac{\cos 4\theta}{\cos 8\theta}$$

$$\Rightarrow \frac{(2\sin 4\theta \cos 4\theta)}{\cos 8\theta} \times \frac{\sin 4\theta}{2\sin^2 2\theta}$$

$$\Rightarrow \frac{\sin 8\theta}{\cos 8\theta} \times \frac{2\sin 2\theta \cos 2\theta}{2\sin^2 2\theta} = \frac{\tan 8\theta}{\tan 2\theta}$$

Hence proved.

Q2 **Prove the identity** $\dfrac{3-4\cos 2A+\cos 4A}{3+4\cos 2A+\cos 4A}=\tan^4 A$

Sol. LHS $\dfrac{4(1-\cos 2A)-(1-\cos 4A)}{4(1+\cos 2A)-(1-\cos 4A)}$

$$\Rightarrow \frac{4\times 2\sin^2 A-2\sin^2 2A}{4\times 2\cos^2 A-2\sin^2 2A}$$

$$\Rightarrow \frac{4\sin^2 A-4\sin^2 A\cos^2 A}{4\cos^2 A-4\sin^2 A\cos^2 A}$$

$$\Rightarrow \frac{\sin^2 A(1-\cos^2 A)}{\cos^2 A(1-\sin^2 A)}=\frac{\sin^4 A}{\cos^4 A}=\tan^4 A = \text{RHS}$$ **Hence proved.**

Q3 **Prove that** $\cos^4\dfrac{\pi}{8}+\cos^4\dfrac{3\pi}{8}+\cos^4\dfrac{5\pi}{8}+\cos^4\dfrac{7\pi}{8}=\dfrac{3}{2}$

Sol. $\cos\dfrac{7\pi}{8}=\cos\left(\pi-\dfrac{\pi}{8}\right)=-\cos\dfrac{\pi}{8}$

$\cos\dfrac{5\pi}{8}=\cos\left(\pi-\dfrac{3\pi}{8}\right)=-\cos\dfrac{3\pi}{8}$

$\therefore$ given expression

LHS $2\cos^4\dfrac{\pi}{8}+2\cos^4\dfrac{3\pi}{8}$

$$\Rightarrow 2\left\{\cos^4\frac{\pi}{8}+\sin^4\left(\frac{\pi}{2}-\frac{3\pi}{8}\right)\right\}$$

$$\Rightarrow 2\left\{\cos^4\frac{\pi}{8}+\sin^4\frac{\pi}{8}\right\}$$

$$\Rightarrow 2\left[\left\{\sin^2\left(\frac{\pi}{8}\right)+\cos^2\left(\frac{\pi}{8}\right)\right\}^2-2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right]$$

$$\Rightarrow 2\left[1-\frac{1}{2}\left(2\sin\frac{\pi}{8}\cos\frac{\pi}{8}\right)^2\right]$$

$$\Rightarrow 2\left[1-\frac{1}{2}\left(\sin\frac{\pi}{4}\right)^2\right]\Rightarrow 2\left[1-\frac{1}{2}\times\frac{1}{2}\right]$$

$$\Rightarrow 2\left(\frac{3}{4}\right)=\frac{3}{2}$$ **Hence proved.**

Q4 **Prove that $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha = \cot\alpha$**

Sol. LHS $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + \dfrac{8}{\tan 8\alpha}$

$\Rightarrow \tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + \dfrac{8\left(1-\tan^2 4\alpha\right)}{2\tan 4\alpha}$

$\Rightarrow \tan\alpha + 2\tan 2\alpha + 4\left\{\tan 4\alpha + \dfrac{1}{\tan 4\alpha} - \tan 4\alpha\right\}$

$\Rightarrow \tan\alpha + 2\tan 2\alpha + \dfrac{4}{\tan 4\alpha}$

$\Rightarrow \tan\alpha + 2\tan 2\alpha + \dfrac{4\left(1-\tan^2 2\alpha\right)}{2\tan 2\alpha}$

$\Rightarrow \tan\alpha + 2\left\{\tan 2\alpha + \dfrac{1}{\tan 2\alpha} - \tan 2\alpha\right\}$

$\Rightarrow \tan\alpha + \dfrac{2}{\tan 2\alpha} \Rightarrow \tan\alpha + \dfrac{2\left(1-\tan^2\alpha\right)}{2\tan\alpha}$

$\Rightarrow \dfrac{1}{\tan\alpha} = \cot\alpha$ **Hence proved.**

Q5 **Let $A_1, A_2, \ldots A_n$ be the vertices of n-sided regular polygon such that $\dfrac{1}{A_1A_2} = \dfrac{1}{A_1A_3} + \dfrac{1}{A_1A_4}$ find value of n.**

Sol. Let radius of circle = R

$\theta = \dfrac{2\pi}{n}$

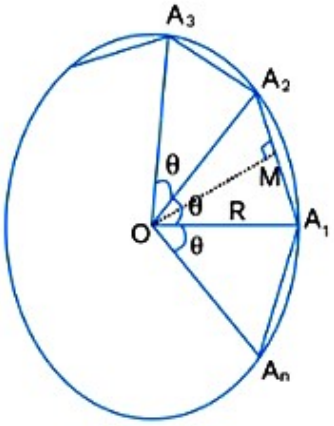
In ΔOMA_1 $MA_1 = R\sin\theta/2$

$\therefore A_1A_2 = 2R\sin\theta/2$

$A_1A_3 = 2R\sin\left(\dfrac{2\theta}{2}\right)$

Similarly $A_1A_4 = 2R\sin\left(\dfrac{3\theta}{2}\right)$

$\because \dfrac{1}{A_1A_2} = \dfrac{1}{A_1A_3} + \dfrac{1}{A_1A_4}$



$$\therefore \frac{1}{2R\sin\theta/2} = \frac{1}{2R\sin(2\theta/2)} + \frac{1}{2R\sin(3\theta/2)}$$

$$\frac{1}{\sin\theta/2} - \frac{1}{\sin 3\theta/2} = \frac{1}{\sin\theta}$$

$$\frac{\sin 3\theta/2 - \sin\theta/2}{\sin\theta/2\sin 3\theta/2} = \frac{1}{\sin\theta}$$

$$\frac{2\cos\theta\sin\theta/2}{\sin\theta/2\sin 3\theta/2} = \frac{1}{\sin\theta}$$

$$\Rightarrow \sin 2\theta = \sin 3\theta/2$$

$$\Rightarrow 2\theta = \pi - 3\theta/2$$

$$\Rightarrow \frac{7\theta}{2} = \pi \Rightarrow \theta = \frac{2\pi}{7} = \frac{2\pi}{n} \Rightarrow n = 7$$

Q6 **Let $\theta \in R$ and $\sum_{k=2}^{\infty} \sin\left(2^k\theta\right) = a$ then find the value of**

$$\sum_{k=0}^{\infty} \left\{\cot^3\left(2^k\theta\right) - \cot\left(2^k\theta\right)\right\} \sin^4\left(2^k\theta\right)$$

Sol. $\left\{\cot^3\alpha - \cot\alpha\right\}\sin^4\alpha$

$$= \left\{\frac{\cos^3\alpha}{\sin^3\alpha} - \frac{\cos\alpha}{\sin\alpha}\right\}\sin^4\alpha$$

$$= \cos^3\alpha\cdot\sin\alpha - \cos\alpha\cdot\sin^3\alpha$$

$$= \sin\alpha\cos\alpha\left\{\cos^2\alpha - \sin^2\alpha\right\}$$

$$= \frac{1}{2}\sin 2\alpha\cdot\{\cos 2\alpha\} = \frac{1}{4}\sin 4\alpha$$

Hence

$$\sum_{k=0}^{\infty} \left\{\cot^3\left(2^k\theta\right) - \cot\left(2^k\theta\right)\right\} \sin^4\left(2^k\theta\right)$$

$$= \frac{1}{4}\sum_{k=0}^{\infty} \sin\left(2^{k+2}\theta\right)$$

$$= \frac{1}{4}\{\sin 2^2\theta + \sin 2^3\theta + \sin 2^4\theta + \cdots)$$

$$= \frac{1}{4}(a)$$

Sine, cosine and tangent of 3A

(i) $\sin 3A = 3\sin A - 4\sin^3 A$

Proof: $\sin 3A = \sin(A + 2A)$

$= \sin A \cos 2A + \cos A \sin 2A$

$= \sin A (1 - 2\sin^2 A) + \cos A \, 2 \sin A \cos A$

$= \sin A - 2\sin^3 A + 2 \sin A (1 - \sin^2 A)$

$= 3\sin A - 4\sin^3 A$

(ii) $\cos 3A = 4\cos^3 A - 3\cos A$

Proof: $\cos 3A = \cos (A + 2A)$

$= \cos A \cos 2A - \sin A \sin 2A$

$= \cos A (2\cos^2 A - 1) - \sin A \, 2\sin A \cos A$

$= 2\cos^3 A - \cos A - 2\cos A(1 - \cos^2 A)$

$= 4 \cos^3 A - 3\cos A$

(iii) $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

Proof: $\tan 3A = \tan\left(A + 2A\right) = \dfrac{\tan A + \tan 2A}{1 - \tan A \tan 2A}$

$$= \frac{\tan A + \dfrac{2\tan A}{1 - \tan^2 A}}{1 - \tan A \cdot \dfrac{2\tan A}{1 - \tan^2 A}}$$

$$= \frac{\tan A\left(1 - \tan^2 A\right) + 2\tan A}{\left(1 - \tan^2 A\right) - 2\tan^2 A} \Rightarrow \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$$

tan(A + B + C)

Proof: $\tan(A + B + C) = \tan\{(A + B) + C\}$

$$= \frac{\tan\left(A + B\right) + \tan C}{1 - \tan\left(A + B\right)\tan C} = \frac{\left\{\dfrac{\tan A + \tan B}{1 - \tan A \tan B}\right\} + \tan C}{1 - \left\{\dfrac{\tan A + \tan B}{1 - \tan A \tan B}\right\}\tan C}$$

On simplifying we get

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

Note: If

(i) $A + B + C = \pi$ then $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

(ii) $A + B + C = \pi/2$ then $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$

Important result:

(i) $\sin\theta \sin(60° - \theta) \sin(60° + \theta) = \frac{1}{4}\sin 3\theta$

Proof: $\sin\theta \{\sin^2 60° - \sin^2\theta\}$

$$\sin\theta\left\{\frac{3}{4} - \sin^2\theta\right\} \Rightarrow \sin\theta\left\{\frac{3 - 4\sin^2\theta}{4}\right\}$$

$$\Rightarrow \frac{1}{4}(3\sin\theta - 4\sin^3\theta\} \Rightarrow \frac{1}{4}\sin 3\theta$$

(ii) $\cos\theta \cos(60° - \theta) \cos(60° + \theta) = \frac{1}{4}\cos 3\theta$

Proof: $\cos\theta\left\{\cos^2\theta - \sin^2 60°\right\}$

$$\cos\theta\left\{\cos^2\theta - \frac{3}{4}\right\} \Rightarrow \cos\theta\left\{\frac{4\cos^2\theta - 3}{4}\right\}$$

$$\Rightarrow \frac{1}{4}\left\{4\cos^3\theta - 3\cos\theta\right\} \Rightarrow \frac{1}{4}\cos 3\theta$$

(iii) $\tan\theta \tan(60° - \theta) \tan(60° + \theta) = \tan 3\theta$

Proof: $\tan\theta\left\{\frac{\tan 60° - \tan\theta}{1 + \tan 60° \tan\theta}\right\}\left\{\frac{\tan 60° + \tan\theta}{1 - \tan 60° \tan\theta}\right\}$

$$\Rightarrow \tan\theta\left\{\frac{\sqrt{3} - \tan\theta}{1 + \sqrt{3}\tan\theta}\right\}\left\{\frac{\sqrt{3} + \tan\theta}{1 - \sqrt{3}\tan\theta}\right\}$$

$$\Rightarrow \frac{\tan\theta\left\{3 - \tan^2\theta\right\}}{1 - 3\tan^2\theta} = \tan 3\theta$$

(iv) $\tan x + \tan(60° + x) - \tan(60° - x) = 3\tan 3x$

Proof: $\tan x + \frac{\sqrt{3} + \tan x}{1 - \sqrt{3}\tan x} - \frac{\sqrt{3} - \tan x}{1 + \sqrt{3}\tan x}$

$\Rightarrow$ Let $\tan x = t$

then $t + \frac{\sqrt{3} + t}{1 - \sqrt{3}t} - \frac{\sqrt{3} - t}{1 + \sqrt{3}t}$

$$\Rightarrow t + \frac{\left(\sqrt{3} + t\right)\left(1 + \sqrt{3}t\right) - \left(\sqrt{3} - t\right)\left(1 - \sqrt{3}t\right)}{\left(1 - \sqrt{3}t\right)\left(1 + \sqrt{3}t\right)}$$

$$\Rightarrow t + \frac{8t}{1-3t^2} = \frac{3\left(3t - t^3\right)}{1-3t^2}$$

$$\Rightarrow \frac{3\left(3\tan\theta - \tan^3\theta\right)}{1-3\tan^2\theta} = 3\tan 3\theta$$

(v) $\cot x + \cot\left(60° + x\right) - \cot\left(60° - x\right) = 3\cot 3x$

Proof: $\cot x + \dfrac{1/\sqrt{3}\cot x - 1}{\cot x + 1/\sqrt{3}} - \dfrac{1/\sqrt{3}\cot x + 1}{\cot x - 1/\sqrt{3}}$

$\Rightarrow \cot x = t$

then $t + \dfrac{t - \sqrt{3}}{\sqrt{3}t + 1} - \dfrac{t + \sqrt{3}}{\sqrt{3}t - 1}$

$$t + \frac{\left(-8t\right)}{3t^2 - 1} = \frac{3t^3 - 9t}{3t^2 - 1}$$

$$\Rightarrow 3\left\{\frac{\cot^3 x - 3\cot x}{3\cot^2 x - 1}\right\} = 3\cot 3x$$

Q1 **Find the value of $8\sin^3 40° - 6\sin 40°$**

Sol. $-2\{3\sin 40° - 4\sin^3 40°\}$

$-2\{\sin 3(40°)\}$

$-2\sin 120° \Rightarrow -2 \times \dfrac{\sqrt{3}}{2} = -\sqrt{3}$

Q2 **Prove that $\dfrac{\sin 3A}{\sin A} - \dfrac{\cos 3A}{\cos A} = 2$**

Sol. $\dfrac{3\sin A - 4\sin^3 A}{\sin A} - \dfrac{4\cos^3 A - 3\cos A}{\cos A}$

$\Rightarrow (3 - 4\sin^2 A) - (4\cos^2 A - 3)$

$\Rightarrow 6 - 4(\sin^2 A + \cos^2 A)$

$\Rightarrow 6 - 4 = 2$

Q3 **Prove that** $\frac{\cos^3\alpha - \cos 3\alpha}{\cos\alpha} + \frac{\sin^3\alpha + \sin 3\alpha}{\sin\alpha} = 3$

Sol. $\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha$

$\cos 3\alpha = \cos^3\alpha + (3\cos^3\alpha - 3\cos\alpha)$

$\frac{\cos^3\alpha - \cos 3\alpha}{\cos\alpha} = 3(1 - \cos^2\alpha) = 3\sin^2\alpha$ .....(i)

$\sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha$

$\sin 3\alpha = (3\sin\alpha - 3\sin^3\alpha) - \sin^3\alpha$

$\frac{\sin^3\alpha + \sin 3\alpha}{\sin\alpha} = 3(1 - \sin^2\alpha) = 3\cos^2\alpha$ .....(ii)

(i) + (ii)

$3(\sin^2\alpha + \cos^2\alpha) \Rightarrow 3$ = RHS **Hence proved.**

Q4 **Prove that** $(4\cos^2 9° - 3)(4\cos^2 27° - 3) = \tan 9°$

Sol. LHS $\left\{\frac{(4\cos^3 9° - 3\cos 9°)}{\cos 9°}\right\}\left\{\frac{4\cos^3 27° - 3\cos 27°}{\cos 27°}\right\}$

$\Rightarrow \frac{\cos 27°}{\cos 9°} \times \frac{\cos 81°}{\cos 27°} \Rightarrow \frac{\cos 81°}{\cos 9°} = \frac{\sin 9°}{\cos 9°} = \tan 9°$

Q5 **To express $\cos 5\theta$ in terms of $\cos\theta$**

Sol. $\cos(5\theta) = \cos(3\theta + 2\theta)$

$= \cos 3\theta\cos 2\theta - \sin 3\theta\sin 2\theta$

$= (4\cos^3\theta - 3\cos\theta)(2\cos^2\theta - 1) - (3\sin\theta - 4\sin^3\theta)(2\sin\theta\cos\theta)$

$= (8\cos^5\theta - 10\cos^3\theta + 3\cos\theta) - (3 - 4\sin^2\theta) \times (2\sin^2\theta \cdot \cos\theta)$

let $\cos\theta = t$

$= (8t^5 - 10t^3 + 3t) - \{3 - 4(1 - t^2)\}\{2(1 - t^2)t\}$

$= (8t^5 - 10t^3 + 3t) - (4t^2 - 1)(2t - 2t^3)$

$= 16t^5 - 20t^3 + 5t$

$\therefore \cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

Q6 **To express sin5θ in terms of sinθ**

Sol. In formula of cos5θ replace θ by $\frac{\pi}{2} - \theta$

$$\cos\left\{5\left(\frac{\pi}{2}-\theta\right)\right\} = 16\cos^5\left(\frac{\pi}{2}-\theta\right) - 20\cos^3\left(\frac{\pi}{2}-\theta\right) + 5\cos\left(\frac{\pi}{2}-\theta\right)$$

$$\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$$

Q7 **Find the value of sin25° sin35° sin85°**

Sol. $\sin 25° \sin(60°-25°) \sin(60°+25°)$

$$= \frac{1}{4}\sin\left\{3(25°)\right\}$$

$$= \frac{1}{4}\sin 75° = \frac{1}{4}\left\{\frac{\sqrt{3}+1}{2\sqrt{2}}\right\} = \frac{\sqrt{3}+1}{8\sqrt{2}}$$

Q8 **Prove that $\cos 6A = 32\cos^6 A - 48\cos^4 A + 18\cos^2 A - 1$**

Sol. $\cos(6A) = \cos\left\{2(3A)\right\}$

$= 2\cos^2(3A) - 1$

$= 2(4\cos^3 A - 3\cos A)^2 - 1$

let $\cos A = t$

$= 2(4t^3 - 3t)^2 - 1$

$= 2(16t^6 - 24t^4 + 9t^2) - 1$

$= 32t^6 - 48t^4 + 18t^2 - 1$

$= 32\cos^6 A - 48\cos^4 A + 18\cos^2 A - 1$

To Play with cos 2θ, sin 2θ, 1 ± cos 2θ, 1 ± sin 2θ

1. $\frac{1+\cos 2\theta}{\sin 2\theta} = \frac{2\cos^2\theta}{2\sin\theta\cos\theta} = \cot\theta$

2. $\frac{1-\cos 2\theta}{\sin 2\theta} = \frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \tan\theta$

3. $\frac{1-\cos 2\theta}{1+\cos 2\theta} = \frac{2\sin^2\theta}{2\cos^2\theta} = \tan^2\theta$

4. To prove that $\frac{1+\sin 2A}{\cos 2A} = \tan(45° + A)$

Proof: LHS $\frac{(\sin A + \cos A)^2}{(\cos^2 A - \sin^2 A)}$

$$\Rightarrow \frac{(\sin A + \cos A)^2}{(\cos A + \sin A)(\cos A - \sin A)} = \frac{\cos A + \sin A}{\cos A - \sin A}$$

$$\Rightarrow \frac{\cos A(1 + \tan A)}{\cos A(1 - \tan A)} = \tan(45° + A)$$

Q1 **Prove that $\tan 7.5° = \cot 82.5° = (\sqrt{3} - \sqrt{2})(\sqrt{2} - 1)$**

Sol. Let $x = 7.5°$

$2x = 15° \Rightarrow \tan 2x = \tan 15°$

$$\Rightarrow \frac{2\tan x}{1 - \tan^2 x} = 2 - \sqrt{3}$$

$$(2 - \sqrt{3})\tan^2 x + 2\tan x - (2 - \sqrt{3}) = 0$$

$$\Rightarrow \tan x = \frac{-2 \pm \sqrt{4 + 4(2 - \sqrt{3})^2}}{2(2 - \sqrt{3})}$$

$$\Rightarrow \tan x = \frac{-1 \pm \sqrt{1 + (7 - 4\sqrt{3})}}{(2 - \sqrt{3})}$$

$$\Rightarrow \tan x = \frac{-1 + \sqrt{8 - 4\sqrt{3}}}{(2 - \sqrt{3})} \quad (\because \tan x > 0)$$

$$\Rightarrow \tan x = \frac{\sqrt{2(\sqrt{3} - 1)^2} - 1}{(2 - \sqrt{3})} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}}$$

$$\Rightarrow \tan x = \{\sqrt{2}(\sqrt{3} - 1) - 1\}\{2 + \sqrt{3}\}$$

$$\Rightarrow \quad \tan x = 2\sqrt{2}\sqrt{3} + 3\sqrt{2} - 2\sqrt{2} - \sqrt{2}\sqrt{3} - 2 - \sqrt{3}$$

$$= \sqrt{2}\sqrt{3} + \sqrt{2} - 2 - \sqrt{3}$$

$$= \sqrt{2}\left(\sqrt{3} - \sqrt{2}\right) - \left(\sqrt{3} - \sqrt{2}\right)$$

$$\tan 7.5^\circ = \left(\sqrt{3} - \sqrt{2}\right)\left(\sqrt{2} - 1\right)$$

Hence proved.

Note:

$$\cot 7.5^\circ = \frac{1}{\left(\sqrt{3} - \sqrt{2}\right)\left(\sqrt{2} - 1\right)} = \left(\sqrt{3} + \sqrt{2}\right)\left(\sqrt{2} + 1\right)$$

Π Notation: Π is used to denote continued product.

$$\prod_{r=1}^{n} \sin\left(2r - 1\right)\theta = \sin\theta.\ \sin 3\theta.\ \sin 5\theta \ldots \sin\left(2n - 1\right)\theta$$

Continued product of cosine series:

Continued product of cosine series in which each angle is double of the previous angle

$$\cos A \cdot \cos 2A \cdot \cos 2^2 A \ldots \cos 2^{n-1} A = \frac{\sin\left(2^n A\right)}{2^n \sin A}$$

Note: If continued product of cosine series is given such that each angle is double of previous angle, not necessarily the last one then multiply and divide the series by 2 sine of smallest angle.

Proof: LHS

$$\frac{2\sin A}{2\sin A} \cdot \cos A.\cos 2A.\ \cos 2^2 A \ldots\ \cos 2^{n-1} A$$

$$\Rightarrow \frac{1}{2^2 \sin A}\left\{2\sin 2A.\ \cos 2A.\cos 2^2 A \ldots \cos 2^{n-1} A\right\}$$

$$\Rightarrow \frac{1}{2^3 \sin A}\left\{2\sin 2^2 A.\ \cos 2^2 A \ldots \cos 2^{n-1} A\right\}$$

and so on ... $\Rightarrow \dfrac{\sin\left(2^n A\right)}{2^n \sin A}$

Q1 **Prove that $\cos36^\circ \cos72^\circ = \frac{1}{4}$**

Sol. LHS $\dfrac{\{2\sin36^\circ \cos36^\circ\}\cos72^\circ}{2\sin36^\circ}$

$\Rightarrow \quad \dfrac{\{2\sin72^\circ \cos72^\circ\}}{4\sin36^\circ} \Rightarrow \dfrac{\sin(144^\circ)}{4\sin36^\circ}$

$\Rightarrow \quad \dfrac{\sin(180^\circ - 36^\circ)}{4\sin36^\circ} = \dfrac{\sin36^\circ}{4\sin36^\circ} = \dfrac{1}{4}$ **Hence proved.**

Q2 **Prove that $\cos20^\circ \cos40^\circ \cos60^\circ \cos80^\circ = \frac{1}{16}$**

Sol. LHS let $A = 20^\circ$

$\cos60^\circ \{\cos A \cos2A \cos2^2A\}$

$\Rightarrow \quad \dfrac{1}{2}\left\{\dfrac{\sin2^3A}{2^3\sin A}\right\} = \dfrac{1}{2}\left\{\dfrac{\sin160^\circ}{8\sin20^\circ}\right\}$

$\Rightarrow \quad \dfrac{1}{16}\dfrac{\sin(180^\circ - 20^\circ)}{\sin20^\circ} \Rightarrow \dfrac{1}{16}$ **Hence proved.**

Q3 **Find value of $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$**

Sol. LHS $\cos\dfrac{\pi}{7}\cos\dfrac{2\pi}{7}\cos\left(\pi - \dfrac{4\pi}{7}\right)$

$\Rightarrow \quad -\cos\dfrac{\pi}{7} \cdot \cos\dfrac{2\pi}{7} \cdot \cos2^2\dfrac{\pi}{7}$

$\Rightarrow \quad -\dfrac{\sin(2^3\pi/7)}{2^3\sin\pi/7} = -\dfrac{\sin(\pi + \pi/7)}{8\sin(\pi/7)}$

$\Rightarrow \quad \dfrac{\sin(\pi/7)}{8\sin(\pi/7)} = \dfrac{1}{8}$

Q4 **Find the value of $\cos\frac{\pi}{10}\cos\frac{2\pi}{10}\cos\frac{4\pi}{10}\cos\frac{8\pi}{10}\cos\frac{16\pi}{10}$**

Sol.

$$\frac{\sin\left(\frac{32\pi}{10}\right)}{32\sin(\pi/10)} = \frac{\sin(3\pi + 2\pi/10)}{32\sin(\pi/10)}$$

$$\Rightarrow \quad \frac{-\sin(2\pi/10)}{32\sin(\pi/10)} \Rightarrow \frac{-2\sin\pi/10\cos\pi/10}{32\sin\pi/10}$$

$$\Rightarrow \quad \frac{-1}{16}\cos\frac{\pi}{10}$$

Q5 **Find value of sin6° sin42° sin66° sin78° or cos12° cos24° cos48° cos84°**

Sol. $\cos84° = \cos(180° - 96°) = -\cos96°$

$\therefore$ $-\cos12°\cos24°\cos48°\cos96°$

$-\cos A\cos2A\cos2^2A\cos2^3A \qquad (A = 12°)$

$$\Rightarrow \quad \frac{-\sin\left(2^4A\right)}{2^4\sin A} \Rightarrow \frac{-\sin(192°)}{16\sin12°}$$

$$\Rightarrow \quad \frac{-\sin\left(180° + 12°\right)}{16\sin12°} = \frac{1}{16}$$

Q6 **Find value of $\prod_{r=1}^{7}\sin\left(2r-1\right)\frac{\pi}{14}$**

Sol. Given expression is

$$\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\sin\frac{7\pi}{14}\sin\frac{9\pi}{14}\sin\frac{11\pi}{14}\sin\frac{13\pi}{14}$$

$$\because \sin\frac{13\pi}{14} = \sin\left(\pi - \frac{\pi}{14}\right) = \sin\frac{\pi}{14}$$

$$\sin\frac{11\pi}{14} = \sin\left(\pi - \frac{3\pi}{14}\right) = \sin\frac{3\pi}{14}$$

$$\sin\frac{9\pi}{14} = \sin\left(\pi - \frac{5\pi}{14}\right) = \sin\frac{5\pi}{14}$$

$\therefore$ given expression

$$\left\{\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\right\}^2 . \sin\frac{7\pi}{14}$$

$$\Rightarrow \left\{\cos\left(\frac{\pi}{2} - \frac{\pi}{14}\right)\cos\left(\frac{\pi}{2} - \frac{3\pi}{14}\right)\cos\left(\frac{\pi}{2} - \frac{5\pi}{14}\right)\right\}^2 \times 1$$

$$\Rightarrow \left\{\cos\frac{3\pi}{7}.\cos\frac{2\pi}{7}.\cos\frac{\pi}{7}\right\}^2$$

$$\Rightarrow \left\{-\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\right\}^2$$

$$\Rightarrow \left\{\frac{-\sin\frac{8\pi}{7}}{8\sin\frac{\pi}{7}}\right\}^2 \Rightarrow \left\{\frac{-\sin\left(\pi + \frac{\pi}{7}\right)}{8\sin\frac{\pi}{7}}\right\}^2$$

$$\Rightarrow \frac{1}{8^2} = \frac{1}{64}$$

Q7 **Find the value of $\sin\frac{\pi}{7}.\ \sin\frac{2\pi}{7}.\ \sin\frac{3\pi}{7}$**

Sol. Let $x = \sin\frac{\pi}{7}\sin\frac{2\pi}{7}.\sin\frac{3\pi}{7}$

$$x^2 = \sin^2\frac{\pi}{7}.\ \sin^2\frac{2\pi}{7}.\ \sin^2\frac{3\pi}{7}$$

$$x^2 = \frac{1}{8}\left(1 - \cos\frac{2\pi}{7}\right)\left(1 - \cos\frac{4\pi}{7}\right)\left(1 - \cos\frac{6\pi}{7}\right)$$

$$x^2 = \frac{1}{8}\left(1 - \cos\frac{2\pi}{7}\right)\left(1 - \cos\frac{4\pi}{7}\right)\left(1 + \cos\frac{\pi}{7}\right) \qquad \text{...(i)}$$

$$\left\{\because \cos\frac{6\pi}{7} = \cos\left(\pi - \frac{\pi}{7}\right) = -\cos\frac{\pi}{7}\right\}$$

let $\cos\frac{\pi}{7} = a,\ \cos\frac{2\pi}{7} = b,\ \cos\frac{4\pi}{7} = c = -\cos\frac{3\pi}{7}$

from (i)

$$x^2 = \frac{1}{8}(1+a)(1-b)(1-c)$$

$$= \frac{1}{8}\{1+(a-b-c)+(-ab-ac+bc)+abc\} \quad ...(ii)$$

$$-ab-ac+bc = \frac{1}{2}\{-2ab-2ac+2bc\}$$

$$= \frac{1}{2}\left\{-2\cos\frac{\pi}{7}\cos\frac{2\pi}{7} - 2\cos\frac{\pi}{7}\cos\frac{4\pi}{7} + 2\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\right\}$$

$$= \frac{1}{2}\left\{-\cos\frac{3\pi}{7} - \cos\frac{\pi}{7} - \cos\frac{5\pi}{7} - \cos\frac{3\pi}{7} + \cos\frac{6\pi}{7} + \cos\frac{2\pi}{7}\right\}$$

$$= \left\{-\cos\frac{\pi}{7} + \cos\frac{2\pi}{7} - \cos\frac{3\pi}{7}\right\}$$

$$\therefore -ab-ac+bc = (-a+b+c)$$

$$\therefore (a-b-c)+(-ab-ac+bc) = 0 \quad ...(iii)$$

Now $abc = \cos\frac{\pi}{7} \cdot \cos\frac{2\pi}{7} \cdot \cos\frac{4\pi}{7}$

$$= \frac{\sin\left(\frac{8\pi}{7}\right)}{8\sin\frac{\pi}{7}} = \frac{\sin\left(\pi+\frac{\pi}{7}\right)}{8\sin\frac{\pi}{7}} = \frac{-1}{8}$$

$$\therefore abc = \frac{-1}{8} \quad ...(iv)$$

Now put values from (iii) and (iv) into (ii)

$$x^2 = \frac{1}{8}\left\{1+0-\frac{1}{8}\right\}$$

$$\Rightarrow x^2 = \frac{1}{8}\cdot\frac{7}{8}$$

$$\Rightarrow x = \frac{\sqrt{7}}{8}$$

$$\therefore \sin\frac{\pi}{7} \cdot \sin\frac{2\pi}{7} \cdot \sin\frac{3\pi}{7} = \frac{\sqrt{7}}{8}$$

To find the value of sin18°, cos36°

Value of sin18°

Let $x = 18°$

$5x = 90° \Rightarrow 3x = 90° - 2x$

$\sin(3x) = \sin(90° - 2x) = \cos 2x$

$\Rightarrow 3\sin x - 4\sin^3 x = 1 - 2\sin^2 x$

$\Rightarrow 4\sin^3 x - 2\sin^2 x - 3\sin x + 1 = 0$

Now let $\sin x = t$

$\Rightarrow 4t^3 - 2t^2 - 3t + 1 = 0$

$\Rightarrow (t-1)(4t^2 + 2t - 1) = 0$

$4t^2 + 2t - 1 = 0 \qquad (\because t \neq 1)$

$$t = \frac{-2 \pm \sqrt{4 + 4(4)}}{2(4)} \Rightarrow t = \frac{-1 \pm \sqrt{5}}{4}$$

$\because x \in$ I quadrant $\Rightarrow t > 0$

Hence $\sin 18° = \dfrac{-1+\sqrt{5}}{4}$

Value of cos18°

$$\cos 18° = \sqrt{1 - \sin^2 18°}$$

$$= \sqrt{1 - \left(\frac{-1+\sqrt{5}}{4}\right)^2}$$

$$= \sqrt{1 - \frac{6 - 2\sqrt{5}}{16}} = \frac{1}{4}\sqrt{10 - 2\sqrt{5}}$$

Hence $\cos 18° = \dfrac{1}{4}\sqrt{10 - 2\sqrt{5}}$

Value of cos36°

$\cos 36° = 1 - 2\sin^2 18°$

$$= 1 - 2\left(\frac{-1+\sqrt{5}}{4}\right)^2$$

$$= 1 - \left(\frac{6 - 2\sqrt{5}}{8}\right) \Rightarrow \frac{2 + 2\sqrt{5}}{8}$$

Hence $\cos 36° = \dfrac{1+\sqrt{5}}{4}$

Value of sin36°

$$\sin 36° = \sqrt{1-\cos^2 36°}$$

$$= \sqrt{1-\left(\frac{1+\sqrt{5}}{4}\right)^2}$$

$$= \sqrt{1-\left(\frac{6+2\sqrt{5}}{16}\right)} \Rightarrow \frac{1}{4}\sqrt{10-2\sqrt{5}}$$

Hence $\sin 36° = \frac{1}{4}\sqrt{10-2\sqrt{5}}$

Note:

(i) $\sin 54° = \cos 36° = \frac{\sqrt{5}+1}{4}$

(ii) $\cos 54° = \sin 36° = \frac{1}{4}\sqrt{10-2\sqrt{5}}$

Q1 **Find the value of sin 132° sin 12°**

Sol. $\frac{1}{2}\{2\sin 132° \sin 12°\}$

$\Rightarrow \frac{1}{2}\{\cos 120° - \cos 144°\}$

$\Rightarrow \frac{1}{2}\left\{\frac{-1}{2} - \cos(180° - 36°)\right\}$

$\Rightarrow \frac{1}{2}\left\{\frac{-1}{2} + \cos 36°\right\} = \frac{1}{2}\left\{\frac{-1}{2} + \frac{\sqrt{5}+1}{4}\right\}$

$\Rightarrow \frac{\sqrt{5}-1}{8}$

Q2 **Find the value of $\cos^2 48° - \sin^2 12°$**

Sol. $\cos(48° + 12°)\cos(48° - 12°)$

$\Rightarrow \cos 60° \cos 36°$

$\Rightarrow \frac{1}{2}\left\{\frac{\sqrt{5}+1}{4}\right\} = \frac{\sqrt{5}+1}{8}$

Q3 **Prove that $\sin\frac{\pi}{5}.\sin\frac{2\pi}{5}.\sin\frac{3\pi}{5}.\sin\frac{4\pi}{5}=\frac{5}{16}$**

Sol. LHS $\sin\frac{\pi}{5}\ \sin\frac{2\pi}{5}\ \sin\left(\pi-\frac{2\pi}{5}\right)\sin\left(\pi-\frac{\pi}{5}\right)$

$$\Rightarrow \sin^2\frac{\pi}{5}\sin^2\frac{2\pi}{5}\Rightarrow\left(1-\cos^2\frac{\pi}{5}\right)\left(1-\cos^2\frac{2\pi}{5}\right)$$

$$\Rightarrow\left(1-\cos^2 36^\circ\right)\left(1-\cos^2 72^\circ\right)$$

$$\Rightarrow\left\{1-\left(\frac{\sqrt{5}+1}{4}\right)^2\right\}\left\{1-\left(\frac{\sqrt{5}-1}{4}\right)^2\right\}$$

$$\Rightarrow\left\{1-\frac{\left(3+\sqrt{5}\right)}{8}\right\}\left\{1-\frac{(3-\sqrt{5})}{8}\right\}$$

$$\Rightarrow\left(\frac{5-\sqrt{5}}{8}\right)\left(\frac{5+\sqrt{5}}{8}\right)\Rightarrow\frac{20}{64}\Rightarrow\frac{5}{16}$$

Hence proved.

Q4 **Find the value of $4\cos\frac{\pi}{10}-3\sec\frac{\pi}{10}-2\tan\frac{\pi}{10}$.**

Sol. $$4\cos\frac{\pi}{10}-\frac{3}{\cos\pi/10}-2\frac{\sin\pi/10}{\cos\pi/10}$$

$$\Rightarrow\frac{4\cos^2\pi/10-3-2\sin\pi/10}{\cos\pi/10}$$

Multiply and divide by $\cos\frac{\pi}{10}$

$$\Rightarrow\frac{4\cos^3\pi/10-3\cos\pi/10-2\sin\pi/10\ \cos\pi/10}{\cos^2\pi/10}$$

$$\Rightarrow\frac{\cos 3\pi/10-\sin 2\pi/10}{\cos^2\pi/10}$$

$$\Rightarrow\frac{\sin\left(\frac{\pi}{2}-\frac{3\pi}{10}\right)-\sin 2\pi/10}{\cos^2\pi/10}=0$$

Conditional Trigonometric Identities:

Q1 **If $A + B + C = \pi$ then prove that $\sin2A + \sin2B + \sin2C = 4\sin A \sin B \sin C$ or $\Sigma\sin2A = 4\Pi\sin A$**

Sol. LHS $\sin2A + \sin2B + \sin2C$

$\Rightarrow 2\sin(A + B)\cos(A - B) + 2\sin C\cos C$

$\Rightarrow 2\sin(\pi - C)\cos(A - B) + 2\sin C\cos(\pi - (A + B))$

$\Rightarrow 2\sin C\cos(A - B) - 2\sin C\cos(A + B)$

$\Rightarrow 2\sin C\{\cos(A - B) - \cos(A + B)\}$

$\Rightarrow 2\sin C(2\sin A\sin B)$

$\Rightarrow 4\sin A\sin B\sin C = RHS$ **Hence proved.**

Q2 **If α, β and γ are the angles of a triangle then show that**

$$\frac{\cos\alpha}{\sin\beta\sin\gamma} + \frac{\cos\beta}{\sin\alpha\sin\gamma} + \frac{\cos\gamma}{\sin\alpha\sin\beta} = 2$$

Sol. LHS $\dfrac{\sin\alpha\cos\alpha + \sin\beta\cos\beta + \sin\gamma\cos\gamma}{\sin\alpha\sin\beta\sin\gamma}$

$$\Rightarrow \frac{1}{2\sin\alpha\sin\beta\sin\gamma}(\sin2\alpha + \sin2\beta + \sin2\gamma)$$

$$\Rightarrow \frac{1}{2\sin\alpha\sin\beta\sin\gamma}(4\sin\alpha\sin\beta\sin\gamma) = 2$$

Q3 **Find the value of** $\dfrac{\sin50° + \sin100° + \sin210°}{\sin25°\sin50°\sin105°}$

Sol. Let $A = 25°$, $B = 50°$, $C = 105°$

$\therefore A + B + C = 180°$

$\therefore \sin2A + \sin2B + \sin2C = 4\sin A\sin B\sin C$

$$\frac{\sin2A + \sin2B + \sin2C}{\sin A\sin B\sin C} = 4$$

$$\Rightarrow \frac{\sin50° + \sin100° + \sin210°}{\sin25°\sin50°\sin105°} = 4$$

Q4 **If $A + B + C = \pi$ then prove that**
$\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A \cos B \cos C$

Sol. $2\cos(A + B)\cos(A - B) + \cos 2C$

$\Rightarrow 2\cos(\pi - C)\cos(A - B) + 2\cos^2 C - 1$

$\Rightarrow 2(-\cos C)\cos(A - B) + 2\cos^2 C - 1$

$\Rightarrow -2\cos C\{\cos(A - B) - \cos C\} - 1$

$\Rightarrow -2\cos C\{\cos(A - B) + \cos(A + B)\} - 1$

$(\because \cos C = \cos\{\pi - (A + B)\} = -\cos(A + B))$

$\Rightarrow -2\cos C(2\cos A \cos B) - 1 = \text{RHS}$ **Hence proved.**

Q5 **If $A + B + C = \pi$ then prove that $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$**

Sol. $\text{LHS } 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\sin\frac{C}{2}\cos\frac{C}{2}$

$\Rightarrow 2\sin\left(\frac{\pi - C}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\sin\frac{C}{2}\cos\frac{C}{2}$

$\Rightarrow 2\cos\frac{C}{2}\left\{\cos\left(\frac{A-B}{2}\right) + \sin\frac{C}{2}\right\}$

$\Rightarrow 2\cos\frac{C}{2}\left\{\cos\left(\frac{A}{2} - \frac{B}{2}\right) + \cos\left(\frac{A}{2} + \frac{B}{2}\right)\right\}$

$\left\{\because \sin\frac{C}{2} = \sin\left(\frac{\pi - (A+B)}{2}\right) = \cos\left(\frac{A+B}{2}\right)\right\}$

$\Rightarrow 2\cos\frac{C}{2}\left\{2\cos\frac{A}{2}\cos\frac{B}{2}\right\}$

$\Rightarrow 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2} = \text{RHS}$ **Hence proved.**

Q6 **If A + B + C = π then prove that $\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$**

Sol. LHS $2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + \cos C$

$\Rightarrow 2\cos\left(\frac{\pi - C}{2}\right)\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2}$

$\Rightarrow 2\sin\frac{C}{2}\left\{\cos\left(\frac{A-B}{2}\right) - \sin\frac{C}{2}\right\} + 1$

$\Rightarrow 2\sin\frac{C}{2}\left\{\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right)\right\} + 1$

$\Rightarrow 2\sin\frac{C}{2}\left\{2\sin\frac{A}{2}\sin\frac{B}{2}\right\} + 1$

$\Rightarrow 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ = RHS **Hence proved.**

Q7 **If A + B + C = π then prove that $\cos^2A + \cos^2B + \cos^2C = 1 - 2\cos A\cos B\cos C$**

Sol. LHS $\cos^2A + (\cos^2B + 1 - \sin^2C)$

$\Rightarrow \cos^2A + (\cos^2B - \sin^2C) + 1$

$\Rightarrow \cos^2A + \cos(B + C)\cos(B - C) + 1$

$\Rightarrow \cos^2A + \cos(\pi - A)\cos(B - C) + 1$

$\Rightarrow \cos^2A - \cos A\cos(B - C) + 1$

$\Rightarrow \cos A\{\cos A - \cos(B - C)\} + 1$

$\Rightarrow \cos A\{\cos\{\pi - (B + C)\} - \cos(B - C)\} + 1$

$\Rightarrow \cos A\{-\cos(B + C) - \cos(B - C)\} + 1$

$\Rightarrow \cos A(-2\cos B\cos C) + 1$

$\Rightarrow 1 - 2\cos A\cos B\cos C$ = RHS **Hence proved.**

Q8 **If A + B + C = π then prove that**

$$\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}=1-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$$

Sol.

$$\text{LHS } \sin^2\frac{A}{2}+\sin\left(\frac{B}{2}+\frac{C}{2}\right)\sin\left(\frac{B}{2}-\frac{C}{2}\right)$$

$$\Rightarrow \sin^2\frac{A}{2}+\sin\left(\frac{\pi}{2}-\frac{A}{2}\right)\sin\left(\frac{B}{2}-\frac{C}{2}\right)$$

$$\Rightarrow 1-\cos^2\frac{A}{2}+\cos\frac{A}{2}\sin\left(\frac{B}{2}-\frac{C}{2}\right)$$

$$\Rightarrow 1-\cos\frac{A}{2}\left\{\cos\frac{A}{2}-\sin\left(\frac{B}{2}-\frac{C}{2}\right)\right\}$$

$$\Rightarrow 1-\cos\frac{A}{2}\left\{\cos\left\{\frac{\pi}{2}-\left(\frac{B+C}{2}\right)\right\}-\sin\left(\frac{B}{2}-\frac{C}{2}\right)\right\}$$

$$\Rightarrow 1-\cos\frac{A}{2}\left\{\sin\left(\frac{B}{2}+\frac{C}{2}\right)-\sin\left(\frac{B}{2}-\frac{C}{2}\right)\right\}$$

$$\Rightarrow 1-\cos\frac{A}{2}\left(2\cos\frac{B}{2}\sin\frac{C}{2}\right)$$

$$\Rightarrow 1-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}=\text{RHS}$$

Hence proved.

Q9 **If A + B + C = π then prove that ΣtanA = ΠtanA i.e., tanA + tanB + tanC = tanA·tanB·tanC**

Sol.

$$A+B+C=\pi$$

$$A+B=\pi-C$$

$$\tan(A+B)=\tan(\pi-C)$$

$$\frac{\tan A+\tan B}{1-\tan A\tan B}=-\tan C$$

$$\Rightarrow \tan A+\tan B=-\tan C+\tan A\tan B\tan C$$

$$\Rightarrow \tan A+\tan B+\tan C=\tan A\tan B\tan C$$

Hence proved.

Q10 **If A + B + C = π then prove that cotA cotB + cotB cotC + cotC cotA = 1 or ΣcotA cotB = 1**

Sol. ∵ ΣtanA = ΠtanA

$$\therefore \frac{1}{\cot A}+\frac{1}{\cot B}+\frac{1}{\cot c}=\frac{1}{\cot A}\cdot\frac{1}{\cot B}\cdot\frac{1}{\cot C}$$

$$\Rightarrow \cot A\cot B+\cot B\cot C+\cot C\cot A=1$$ **Hence proved.**

Q11 **If A + B + C = π then prove that**

$$\tan\frac{A}{2}\tan\frac{B}{2}+\tan\frac{B}{2}\tan\frac{C}{2}+\tan\frac{C}{2}\tan\frac{A}{2}=1$$

Sol. ∵ A + B + C = π

∴ A + B = π − C

$$\frac{A}{2}+\frac{B}{2}=\frac{\pi}{2}-\frac{C}{2}$$

$$\Rightarrow \tan\left(\frac{A}{2}+\frac{B}{2}\right)=\tan\left(\frac{\pi}{2}-\frac{C}{2}\right)$$

$$\Rightarrow \frac{\tan A/2+\tan B/2}{1-\tan A/2\tan B/2}=\cot\frac{C}{2}=\frac{1}{\tan C/2}$$

$$\Rightarrow \tan\frac{A}{2}\tan\frac{C}{2}+\tan\frac{B}{2}\tan\frac{C}{2}=1-\tan\frac{A}{2}\tan\frac{B}{2}$$

$$\Rightarrow \tan\frac{A}{2}\tan\frac{B}{2}+\tan\frac{B}{2}\tan\frac{C}{2}+\tan\frac{C}{2}\tan\frac{A}{2}=1$$ **Hence proved.**

Q12 **In Δ ABC, prove that** $\sum\cot\frac{A}{2}=\Pi\cot\frac{A}{2}$.

i.e., $\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}=\cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}$

Sol. $\therefore \sum\tan\frac{A}{2}\tan\frac{B}{2}=1$

$$\therefore \frac{1}{\cot\frac{A}{2}\cot\frac{B}{2}}+\frac{1}{\cot\frac{B}{2}\cot\frac{C}{2}}+\frac{1}{\cot\frac{C}{2}\cot\frac{A}{2}}=1$$

$$\Rightarrow \cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}=\cot\frac{A}{2}\cdot\cot\frac{B}{2}\cdot\cot\frac{C}{2}$$ **Hence proved.**

Q13 **If A + B + C + D = 360° then prove that**

$$\sin A - \sin B + \sin C - \sin D = -4\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A+D}{2}\right)\sin\left(\frac{A+C}{2}\right)$$

Sol. $\sin A - \sin B + \sin C - \sin D$

$$\Rightarrow 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right) + 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$$

$$\left\{\because \cos\left(\frac{C+D}{2}\right) = \cos\left\{180° - \left(\frac{A+B}{2}\right)\right\} = -\cos\left(\frac{A+B}{2}\right)\right\}$$

$$\Rightarrow 2\cos\left(\frac{A+B}{2}\right)\left\{\sin\left(\frac{A-B}{2}\right) - \sin\left(\frac{C-D}{2}\right)\right\}$$

$$\Rightarrow 2\cos\left(\frac{A+B}{2}\right)\left\{2\cos\left(\frac{A+C-B-D}{4}\right)\sin\left(\frac{A+D-B-C}{4}\right)\right\}$$

$$\Rightarrow 2\cos\left(\frac{A+B}{2}\right)\left\{2\cos\left(\frac{2(A+C)-360°}{4}\right)\sin\left(\frac{2(A+D)-360°}{4}\right)\right\}$$

$$\Rightarrow 2\cos\left(\frac{A+B}{2}\right)\left\{2\sin\left(\frac{A+C}{2}\right)\left(-\cos\left(\frac{A+D}{2}\right)\right)\right\}$$

$$\Rightarrow -4\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A+D}{2}\right)\sin\left(\frac{A+C}{2}\right)$$

Hence proved.

Q14 **If x + y + z = xyz then prove that** $\frac{2x}{1-x^2} + \frac{2y}{1-y^2} + \frac{2z}{1-z^2} = \frac{2x}{1-x^2}\cdot\frac{2y}{1-y^2}\cdot\frac{2z}{1-z^2}$

Sol. Let $x = \tan A$, $y = \tan B$, $z = \tan C$
(where $A + B + C = \pi$) because it satisfies the given relation $x + y + z = xyz$
now $\because A + B + C = \pi$

$$2A + 2B + 2C = 2\pi$$

$$\Rightarrow 2A + 2B = 2\pi - 2C$$

$$\Rightarrow \tan(2A + 2B) = \tan(2\pi - 2C)$$

$$\Rightarrow \frac{\tan 2A + \tan 2B}{1 - \tan 2A \tan 2B} = -\tan 2C$$

$$\Rightarrow \tan 2A + \tan 2B + \tan 2C = \tan 2A \tan 2B \tan 2C$$

$$\text{where } \tan 2A = \frac{2\tan A}{1-\tan^2 A} = \frac{2x}{1-x^2}$$

$$\tan 2B = \frac{2y}{1-y^2}, \tan 2C = \frac{2z}{1-2^2}$$

Hence proved.

Q15 **Number of triangles, tangents of whose angle are 1, 2, 3**
(A) 0 **(B) 1** **(C) ∞** **(D) Nothing can be said**

Sol. **(B)**
Let tanA = 1, tanB = 2, tanC= 3
$\because$ tanA + tanB + tanC = tanA tanB tanC
$\Rightarrow$ Triangle is possible with these values but exactly one values of A, B, C is possible hence only one triangle is possible.

Q16 **If A + B + C = π then prove that**

$$\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}-1=4\sin\left(\frac{\pi-A}{4}\right)\sin\left(\frac{\pi-B}{4}\right)\sin\left(\frac{\pi-c}{4}\right)$$

Sol. LHS $2\sin\left(\frac{A+B}{4}\right)\cos\left(\frac{A-B}{4}\right)+\sin\frac{C}{2}-1$

$$\Rightarrow 2\sin\left(\frac{\pi-C}{4}\right)\cos\left(\frac{A-B}{4}\right)-\left(\cos\frac{C}{4}-\sin\frac{C}{4}\right)^2$$

$$\Rightarrow 2\sin\left(\frac{\pi-C}{4}\right)\cos\left(\frac{A-B}{4}\right)-2\left\{\sin\left(\frac{\pi}{4}-\frac{C}{4}\right)\right\}^2$$

$$\Rightarrow 2\sin\left(\frac{\pi-C}{4}\right)\left\{\cos\left(\frac{A-B}{4}\right)-\sin\left(\frac{\pi-C}{4}\right)\right\}$$

$$\Rightarrow 2\sin\left(\frac{\pi-C}{4}\right)\left\{\cos\left(\frac{A-B}{4}\right)-\sin\left(\frac{A+B}{4}\right)\right\}$$

$$\Rightarrow 2\sin\left(\frac{\pi-C}{4}\right)\left\{\cos\left(\frac{A-B}{4}\right)-\cos\left(\frac{\pi}{2}-\frac{(A+B)}{4}\right)\right\}$$

$$\Rightarrow 2\sin\left(\frac{\pi-C}{4}\right)\left\{2\sin\left(\frac{\pi-B}{4}\right)\sin\left(\frac{\pi-A}{4}\right)\right\}$$

$$\Rightarrow 2\sin\left(\frac{\pi-A}{4}\right)\sin\left(\frac{\pi-B}{4}\right)\sin\left(\frac{\pi-C}{4}\right)=\text{RHS}$$ **Hence proved.**

Q17 **If A + B + C = 2S then prove that 4sinS sin(S – A) sin(S – B) sin(S – C) = 1 – cos²A – cos²B – cos²C + 2 cosA cosB cosC**

Sol. LHS {2sinS. sin(S –A)} {2 sin(S – B) sin(S – C)}
$\Rightarrow$ {cosA – cos(2S – A)} {cos(C – B) – cos(2S – B – C)}
$\Rightarrow$ {cosA – cos(B + C)} {cos(C – B) – cosA}
$\Rightarrow$ cosA {cos(C – B) + cos(B + C)} – cos(B + C) cos(B – C) – cos²A
$\Rightarrow$ cosA (2cosB cosC) – (cos²B – sin²C) – cos²A
$\Rightarrow$ 2cosA cosB cosC – cos²B + 1 – cos²C – cos²A = RHS **Hence proved.**

General formula

$$\tan(\theta_1 + \theta_2 + \theta_3 + \ldots + \theta_n) = \frac{S_1 - S_3 + S_5 - S_7 + \cdots}{1 - S_2 + S_4 - S_6 + \cdots}$$

where S_1 = sum of tangents of angles taken one at a time.

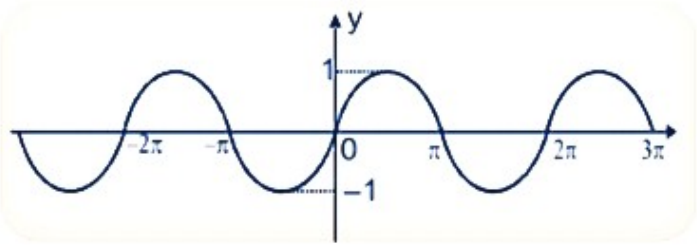
i.e., $\tan\theta_1 + \tan\theta_2 + \ldots + \tan\theta_n$

S_2 = sum of the product of 2 tangents.

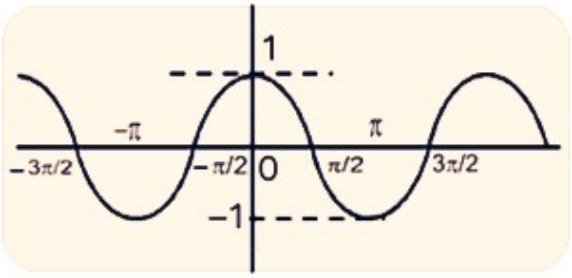
i.e., $\tan\theta_1 \tan\theta_2 + \tan\theta_2 \tan\theta_3 + \tan\theta_3 \tan\theta_4 + \ldots$ etc.

GRAPHS OF 6 TRIGONOMETRIC FUNCTIONS

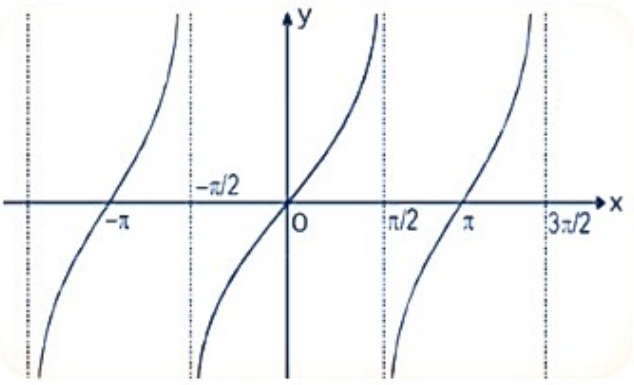
1. $y = \sin x$, where $y \in [-1, 1], x \in R$



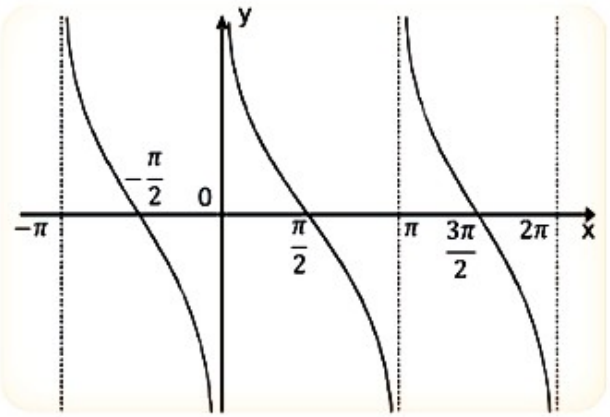
2. $y = \cos x$, where $y \in [-1, 1], x \in R$



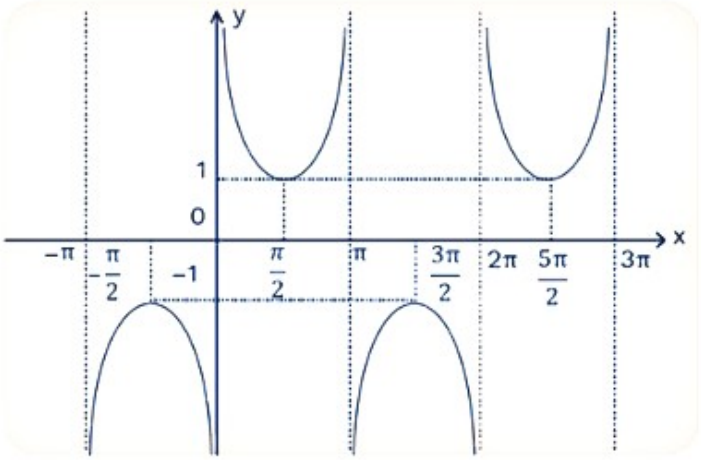
3. $y = \tan x$, where $y \in (-\infty, \infty), x \in R - (2n-1)\frac{\pi}{2}$ for $n \in I$



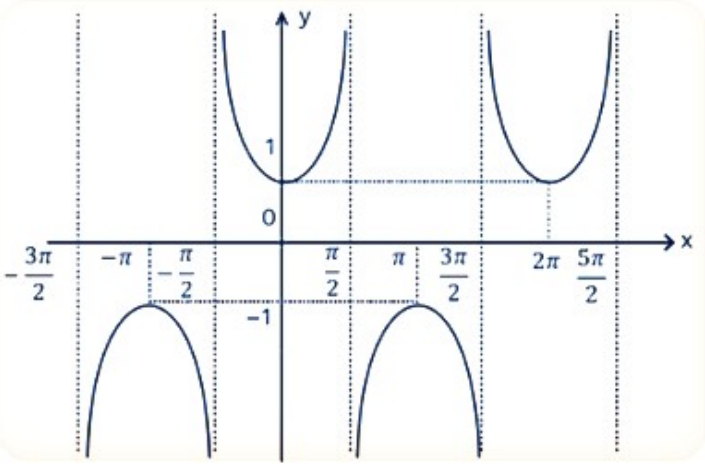
4. $y = \cot x$, where $y \in (-\infty, \infty), x \in R - n\pi$ for $n \in I$



5. $y = \text{cosec } x$, where $y \in (-\infty, -1] \cup [1, \infty), x \in R - n\pi$ for $n \in I$



6. $y = \sec x$, where $y \in (-\infty, -1] \cup [1, \infty), x \in R - (2n - 1)\frac{\pi}{2}$ for $n \in I$



Application of trigonometry in maximising and minimising i.e. (optimisation)

We know that maximising and minimizing by calculus but in certain cases trigonometry proves to be more powerful.

Type-1:

Maximising and minimising by using the property of boundness of trigonometric functions.

(a) Sine and cosine have bounded values between -1 and 1.

(b) Tangent and cotangent are unbounded functions.

(c) Cosec and sec have values greater than or equal to 1 and less than or equal to -1.

Q1 **Find range of $y = 4\tan x \cos x$**

Sol. Given function $y = 4\dfrac{\sin x}{\cos x}\cdot\cos x$

$\Rightarrow y = 4\sin x$ where $x \in R - (2n-1)\dfrac{\pi}{2}$, $n \in I$

$\therefore$ Range of $y \in (-4, 4)$

Q2 **Find range of y where** $y = \cos^4\dfrac{x}{2} - \sin^4\dfrac{x}{2}$

Sol. $y = \left(\cos^2\dfrac{x}{2} + \sin^2\dfrac{x}{2}\right)\left(\cos^2\dfrac{x}{2} - \sin^2\dfrac{x}{2}\right)$

$\Rightarrow y = 1 \times \cos 2\left(\dfrac{x}{2}\right)$

$\Rightarrow y = \cos x$

Hence range of $y \in [-1, 1]$

Q3 **Let $y = \sin\sqrt{x}$. Find the range of $y(x \geq 0)$**

Sol. $\because\ y(x \geq 0) \Rightarrow \sqrt{x} \geq 0$

$\therefore\ \sin\sqrt{x} \in [-1, 1]$

Q4 **$y = \tan^2\left(x - \dfrac{\pi}{2}\right)$. Find range of y.**

Sol. $\because\ \tan x \in R$

$\therefore\ \tan\left(x - \dfrac{\pi}{2}\right) \in R$

$$\Rightarrow \tan^2\left(x-\frac{\pi}{2}\right)\in[0,\infty)$$

Q5 **$y=\sin^2\left(\frac{15\pi}{8}-4x\right)-\sin^2\left(\frac{17\pi}{8}-4x\right)$. Find the range of y.**

Sol.

$$y=\sin\left(\frac{15\pi}{8}-4x+\frac{17\pi}{8}-4x\right)\sin\left(\frac{15\pi}{8}-4x-\frac{17\pi}{8}+4x\right)$$

$$y=\sin(4\pi-8x)\sin\left(-\frac{\pi}{4}\right)$$

$$y=\frac{1}{\sqrt{2}}\sin 8x$$

$$\because \sin 8x\in[-1,1] \quad \therefore \frac{1}{\sqrt{2}}\sin 8x\in\left[\frac{-1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right]$$

Special case

When expression contains terms like (asinx + bcosx) only.

i.e., when argument of sine and cosine are same.

$y = a\cos\theta + b\sin\theta$

Multiply and divide by $\sqrt{a^2+b^2}$

$$y=\sqrt{a^2+b^2}\left\{\frac{a}{\sqrt{a^2+b^2}}\cos\theta+\frac{b}{\sqrt{a^2+b^2}}\sin\theta\right\}$$

Now let $\frac{a}{\sqrt{a^2+b^2}}=\sin\alpha, \frac{b}{\sqrt{a^2+b^2}}=\cos\alpha$

$$y=\sqrt{a^2+b^2}\,(\sin\alpha\cos\theta+\cos\alpha\sin\theta)$$

$$\Rightarrow y=\sqrt{a^2+b^2}\sin(\theta+\alpha)$$

$$\because \sin(\theta+\alpha)\in[-1,1]$$

$$\therefore \sqrt{a^2+b^2}\sin(\theta+\alpha)\in\left[-\sqrt{a^2+b^2},\sqrt{a^2+b^2}\right]$$

$$-\sqrt{a^2+b^2}\le a\cos\theta+b\sin\theta\le\sqrt{a^2+b^2}$$

Q1 $y = \log_2\left(\frac{3\sin x - 4\cos x + 15}{10}\right)$. **Find minimum and maximum value of y.**

Sol. $-\sqrt{3^2 + (-4)^2} \le 3\sin x - 4\cos x \le \sqrt{3^2 + (-4)^2}$

$-5 \le 3\sin x - 4\cos x \le 5$

$\frac{-5+15}{10} \le \frac{3\sin x - 4\cos x + 15}{10} \le \frac{5+15}{10}$

$\log_2 1 \le \log_2\left(\frac{3\sin x - 4\cos x + 15}{10}\right) \le \log_2 2$

Minimum value of $y = \log_2 1 = 0$

Maximum value of $y = \log_2 2 = 1$

Q2 $y = \frac{17 + 5\sin x + 12\cos x}{17 - 5\sin x - 12\cos x}$. **Find the minimum and maximum value of y.**

Sol. $-\sqrt{5^2 + 12^2} \le 5\sin x + 12\cos x \le \sqrt{5^2 + 12^2}$

$-13 \le 5\sin x + 12\cos x \le 13$

then $y = \frac{17+\alpha}{17-\alpha} = -\left\{\frac{\alpha+17}{\alpha-17}\right\}$

$y = -\left\{\frac{(\alpha-17)+34}{\alpha-17}\right\} = -\left\{1 + \frac{34}{\alpha-17}\right\}$

$y = -1 + \frac{34}{17-\alpha}$

$\because -13 \le \alpha \le 13$

$\Rightarrow 4 \le 17 - \alpha \le 30$

$\frac{34}{30} \le \frac{34}{17-\alpha} \le \frac{34}{4}$

$\frac{4}{30} \le -1 + \frac{34}{17-\alpha} \le \frac{30}{4}$

$\frac{2}{15} \le y \le \frac{15}{2}$

$\therefore y_{min} = \frac{2}{15}, \quad y_{max} = \frac{15}{2}$

Q3 **Let f(x) = a cos(α+x) + b cos (β+x). Find maximum and minimum values of f(x), x ∈ R and a, b are constant.**

Sol. $f(x) = a\{\cos\alpha\ \cos x - \sin\alpha\ \sin x\} + b\{\cos\beta\ \cos x - \sin\beta\ \sin x\}$

$\Rightarrow$ f(x) = (acosα + b cosβ) cosx − (a sinα + b sinβ) sinx

Let a cosα + b cosβ = m

a sinα + b sinβ = n

$\therefore$ f(x) = m cos x − n sinx

$\Rightarrow$ $f(x) \in \left[-\sqrt{m^2+n^2},\ \sqrt{m^2+n^2}\right]$

$m^2+n^2=(a\cos\alpha + b\cos\beta)^2 + (a\sin\alpha + b\cos\beta)^2$

$m^2 + n^2 = a^2 + b^2 + 2ab\cos(\alpha - \beta)$

$(m^2 + n^2)_{max} = a^2 + b^2 + 2ab = (a + b)^2$

$\therefore$ $f(x)_{min} = -\sqrt{(m^2+n^2)_{max}} = -|a+b|$

$\therefore$ $f(x)_{max} = \sqrt{(m^2+n^2)_{max}} = |a+b|$

Q4 **$y = \sin\left(x + \frac{\pi}{3}\right) + 3\cos\left(x - \frac{\pi}{3}\right)$. Find minimum and maximum value of y.**

Sol. $y = \left\{\sin x\ \cos\frac{\pi}{3} + \cos x\ \sin\frac{\pi}{3}\right\} + 3\left\{\cos x\ \cos\frac{\pi}{3} + \sin x\ \sin\frac{\pi}{3}\right\}$

$y = \left(\frac{1}{2} + \frac{3\sqrt{3}}{2}\right)\sin x + \left(\frac{\sqrt{3}}{2} + \frac{3}{2}\right)\cos x$

$\left(m = \left(\frac{1}{2} + \frac{3\sqrt{3}}{2}\right),\ n = \left(\frac{\sqrt{3}}{2} + \frac{3}{2}\right)\right)$

$m^2 + n^2 = \left(\frac{1}{4} + \frac{27}{4} + \frac{3\sqrt{3}}{2}\right) + \left(\frac{3}{4} + \frac{9}{4} + \frac{3\sqrt{3}}{2}\right)$

$= 10 + 3\sqrt{3}$

$y_{min} = -\sqrt{m^2+n^2} = -\sqrt{10+3\sqrt{3}}$

$y_{max} = \sqrt{m^2+n^2} = \sqrt{10+3\sqrt{3}}$

Q5 $y = 3\cos\left(x + \frac{\pi}{3}\right) + 5\cos x + 3$ **. Find minimum and maximum value of y.**

Sol. $y = 3\left\{\cos x.\ \cos\frac{\pi}{3} - \sin x\ \sin\frac{\pi}{3}\right\} + 5\cos x + 3$

$$y_{min} = -\sqrt{\left(\frac{13}{2}\right)^2 + \left(\frac{3\sqrt{3}}{2}\right)^2} + 3 = -7 + 3 = -4$$

$$y_{max} = \sqrt{(13/2)^2 + \left(3\sqrt{3}/2\right)^2} + 3 = 7 + 3 = 10$$

Type-2:

Argument of sine and cosine are different or a quadratic in sine / cos is given then we make a perfect square in sine /cosine and interpret.

Q1 $y = \cos 2x + 3\sin x$ **. Find range, minimum and maximum value of y.**

Sol. $y = 1 - 2\sin^2 x + 3\sin x$

$$= -2\left\{\sin^2 x - \frac{3}{2}\sin x\right\} + 1$$

$$= -2\left\{\sin^2 x - \frac{3}{2}\sin x + \frac{9}{16} - \frac{9}{16}\right\} + 1$$

$$= -2\left\{\left(\sin x - \frac{3}{4}\right)^2 - \frac{9}{16}\right\} + 1$$

$$= -2\left(\sin x - \frac{3}{4}\right)^2 + \frac{17}{8}$$

$$y_{min} = -2\left(-1 - \frac{3}{4}\right)^2 + \frac{17}{8} = -4$$

$$y_{max} = -2\left(\frac{3}{4} - \frac{3}{4}\right)^2 + \frac{17}{8} = \frac{17}{8}$$

Q2 **$y = \cos^2 x - 4\cos x + 13$. Find minimum and maximum value of y.**

Sol. $y = \cos^2 x - 4\cos x + 4 + 9$

$= (\cos x - 2)^2 + 9$

$y_{min} = (1-2)^2 + 9 = 10$

$y_{max} = (-1-2)^2 + 9 = 18$

Q3 **If $A = \cos^2\theta + \sin^4\theta$ then find range of A.**

Sol. $A = 1 - \sin^2\theta + \sin^4\theta$

$= \left(\sin^2\theta\right)^2 - \sin^2\theta + \frac{1}{4} + \frac{3}{4}$

$= \left(\sin^2\theta - \frac{1}{2}\right)^2 + \frac{3}{4}$

$A_{min} = \left(\frac{1}{2} - \frac{1}{2}\right)^2 + \frac{3}{4} = \frac{3}{4}$

$A_{max} = \left(1 - \frac{1}{2}\right)^2 + \frac{3}{4} = 1$

$\therefore \frac{3}{4} \le A \le 1$

Type-3:

Making use of reciprocal relationship between tan/cot, sin/cosec and cos/sec.

Q1 **$y = a^2\tan^2 x + b^2\cot^2 x\,(a, b \ge 0)$ Find y_{min}**

Sol. $y = a^2\tan^2 x + \frac{b^2}{\tan^2 x}$

$y = \left(a\tan x - \frac{b}{\tan x}\right)^2 + 2ab$

y will be minimum if $a\tan x - \frac{b}{\tan x} = 0$

$\tan x = \sqrt{b/a}$

Hence $y_{min} = 2ab$

Q2 $y = a^2\sec^2x + b^2\operatorname{cosec}^2x$. Find y_{min}

Sol. $y = a^2\left(1+\tan^2x\right) + b^2\left(1+\cot^2x\right)$

$y = \left(a^2+b^2\right) + a^2\tan^2x + b^2\cot^2x$

$\therefore\ y_{min} = a^2+b^2+\left(a^2\tan^2x + \frac{b^2}{\tan^2x}\right)_{min}$

$y_{min} = a^2+b^2+2ab = \left(a+b\right)^2$

Q3 $y = 4\sin^2x + \operatorname{cosec}^2x$. Find y_{min}

Sol. $y = 4\sin^2x + \frac{1}{\sin^2x}$

$y = \left(2\sin x - \frac{1}{\sin x}\right)^2 + 2\left(2\sin x\right)\left(\frac{1}{\sin x}\right)$

$y = \left(2\sin x - \frac{1}{\sin x}\right)^2 + 4$

y is min. when $2\sin x - \frac{1}{\sin x} = 0$ k $\Rightarrow\ \sin x = \frac{1}{\sqrt{2}}$

Hence $y_{min} = 0 + 4 = \mathbf{4}$

Q4 $y = 8\sec^2x + 18\cos^2x$. Find y_{min}

Sol. $y = \frac{8}{\cos^2 x} + 18\cos^2x$

$y = 2\left\{\frac{4}{\cos^2x} + 9\cos^2x\right\}$

$y = 2\left\{\left(\frac{2}{\cos x} - 3\cos x\right)^2 + \frac{4}{\cos x}3\cos x\right\}$

y is minimum when $\frac{2}{\cos x} - 3\cos x = 0\ \Rightarrow \cos x = \sqrt{\frac{2}{3}}$

Hence $y_{min} = 2\left\{0+12\right\} = \mathbf{24}$

Q5 $y = 18\sec^2 x + 8\cos^2 x$. **Find y_{min}**

Sol. $$y = 2\left\{\frac{9}{\cos^2 x} + 4\cos^2 x\right\}$$

$$y = 2\left\{\left(\frac{3}{\cos x}\right)^2 + (2\cos x)^2\right\}$$

$$y = 2\left\{\left(\frac{3}{\cos x} - 2\cos x\right)^2 + 2.\frac{3}{\cos x}.2\cos x\right\}$$

but in this case $\frac{3}{\cos x} - 2\cos x = 0 \Rightarrow \cos x = \sqrt{\frac{3}{2}}$

Which is not possible hence we write the expression as

$$y = 2\left\{\frac{9}{\cos^2 x} + 9\cos^2 x - 5\cos^2 x\right\}$$

$$y = 2\left\{\left(\frac{3}{\cos x} - 3\cos x\right)^2 + 2.\frac{3}{\cos x}.3\cos x - 5\cos^2 x\right\}$$

now y is minimum if $\frac{3}{\cos x} - 3\cos x = 0 \Rightarrow \cos x = 1$

also in this case $-5\cos^2 x$ is also minimum

Hence $y_{min} = 2\{0 + 18 - 5\}$ ($\because \cos^2 x = 1$)

$\therefore$ $y_{min} = 2(13) = \mathbf{26}$

Type-4:

General or miscellaneous problem

Q1 $\mathbf{x^2 + y^2 = 4}$ **and** $\mathbf{a^2 + b^2 = 8}$

Find (ax + by) minimum and maximum value.

Sol. $\because$ $x^2 + y^2 = 2^2$

$\therefore$ let $x = 2\cos\theta$, $y = 2\sin\theta$

Also $a^2+b^2 = \left(2\sqrt{2}\right)^2$

So let $a = 2\sqrt{2}\cos\alpha, b = 2\sqrt{2}\sin\alpha$

Now $ax + by = 4\sqrt{2}\cos\theta\cos\alpha + 4\sqrt{2}\sin\theta\sin\alpha$

$= 4\sqrt{2}\cos(\theta - \alpha)$

$(ax + by)_{min} = -4\sqrt{2}, (ax + by)_{max} = 4\sqrt{2}$

Q2 $\mathbf{y = \frac{sin3x}{sinx}}$**, find range of y.**

Sol. $y = \frac{3\sin x - 4\sin^3 x}{\sin x} = 3 - 4\sin^2 x$

$0 \le \sin^2 x \le 1$

$0 \ge -\sin^2 x \ge -1$

$3 \ge 3 - 4\sin^2 x \ge -1$

$\therefore$ **range** $\in$ **[-1, 3]**

Q3 $\mathbf{y = \frac{1 + cos2x + 8sin^2 x}{sin2x}}$**, find** $\mathbf{y_{min}}$ **if** $\mathbf{x \in \left(0, \frac{\pi}{2}\right)}$**.**

Sol. $y = \frac{2\cos^2 x + 8\sin^2 x}{2\sin x\cos x} = \cot x + 4\tan x$

$y = \frac{1}{\tan x} + 4\tan x$

$y = \left\{\frac{1}{\sqrt{\tan x}} - 2\sqrt{\tan x}\right\}^2 + 2\frac{1}{\sqrt{\tan x}}2\sqrt{\tan x}$

y is minimum if $\frac{1}{\sqrt{\tan x}} - 2\sqrt{\tan x} = 0 \Rightarrow \tan x = \frac{1}{2}$

then $y_{min} = 0 + 4 = 4$

Q4 **Given $\sin\theta + \sin\phi = a$ and $\cos\theta + \cos\phi = b$. Find the value of**

(i) $\tan\left(\frac{\theta-\phi}{2}\right)$ **(ii)** $\cos(\theta+\phi)$

Sol. $\sin\theta + \sin\phi = a$ ...(i)

$\cos\theta + \cos\phi = b$ ...(ii)

$(i)^2 + (ii)^2$

$(\sin^2\theta + \cos^2\theta) + (\sin^2\phi + \cos^2\phi) + 2(\cos\theta\cos\phi + \sin\theta\sin\phi) = a^2 + b^2$

$2 + 2\cos(\theta - \phi) = a^2 + b^2$

$$\text{hence } 2\left\{2\cos^2\left(\frac{\theta-\phi}{2}\right)\right\} = a^2 + b^2$$

$$\Rightarrow \cos^2\left(\frac{\theta-\phi}{2}\right) = \frac{a^2+b^2}{4}$$

$$\tan^2\left(\frac{\theta-\phi}{2}\right) = \frac{1}{\cos^2\left(\frac{\theta-\phi}{2}\right)} - 1 = \frac{4}{a^2+b^2} - 1$$

$$\tan\left(\frac{\theta-\phi}{2}\right) = \pm\sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$$

$$\text{from (i)} \Rightarrow 2\sin\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = a \quad \text{...(iii)}$$

$$\text{from (ii)} \Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = b \quad \text{...(iv)}$$

$$\text{(iii)} \div \text{(iv)} \Rightarrow \tan\left(\frac{\theta+\phi}{2}\right) = \frac{a}{b}$$

$$\cos(\theta+\phi) = \frac{1-\tan^2\left(\frac{\theta+\phi}{2}\right)}{1+\tan^2\left(\frac{\theta+\phi}{2}\right)} = \frac{1-a^2/b^2}{1+a^2/b^2}$$

$$\cos(\theta+\phi) = \frac{b^2-a^2}{b^2+a^2}$$

Q5 **Prove that** $\frac{\tan 2^n\theta}{\tan\theta} = (1+\sec 2\theta)(1+\sec 2^2\theta)(1+\sec^3\theta).....(1+\sec 2^n\theta)$

Sol. $\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta} \Rightarrow \sec 2\theta = \frac{1+\tan^2\theta}{1-\tan^2\theta}$

$1+\sec 2\theta = \frac{2}{1-\tan^2\theta} = \frac{2\tan\theta}{1-\tan^2\theta}\times\frac{1}{\tan\theta}$

$(1+\sec 2\theta) = \frac{\tan 2\theta}{\tan\theta}$ ...(i)

Similarly $1+\sec 2^2\theta = \frac{\tan 2^2\theta}{\tan 2\theta}$ ...(ii)

$1+\sec 2^n\theta = \frac{\tan 2^n\theta}{\tan 2^{n-1}\theta}$ ...(n)

Multiplying all equations we get

$(1+\sec 2\theta)(1+\sec 2^2\theta)...(1+\sec 2^n\theta)$

$= \frac{\tan 2\theta}{\tan\theta}\cdot\frac{\tan 2^2\theta}{\tan 2\theta}\cdots\frac{\tan 2^n\theta}{\tan 2^{n-1}\theta} = \frac{\tan 2^n\theta}{\tan\theta}$ **Hence proved.**

Q6 **Find the exact value of** $\sqrt{3}\text{cosec}20° - \sec 20°$

Sol. $\frac{\sqrt{3}}{\sin 20°} - \frac{1}{\cos 20°} \Rightarrow \frac{\sqrt{3}\cos 20° - \sin 20°}{\sin 20°\cos 20°}$

$\Rightarrow 2\left(\frac{\sqrt{3}}{2}\cos 20° - \frac{1}{2}\sin 20°\right)\times\frac{2}{2\sin 20°\cos 20°}$

$\Rightarrow 2(\sin 60°\cos 20° - \cos 60°\sin 20°)\times\frac{2}{\sin 40°}$

$\Rightarrow 2\sin(60° - 20°)\times\frac{2}{\sin 40°} = 4$

Q7 **Find the exact value of cot70° + 4cos70°**

Sol.

$$\frac{\cos 70^\circ}{\sin 70^\circ} + 4\cos 70^\circ \Rightarrow \frac{\sin 20^\circ}{\cos 20^\circ} + 4\sin 20^\circ$$

$$\Rightarrow \frac{\sin 20^\circ + 4\sin 20^\circ \cos 20^\circ}{\cos 20^\circ} \Rightarrow \frac{\sin 20^\circ + 2\sin 40^\circ}{\cos 20^\circ}$$

$$\Rightarrow \frac{(\sin 20^\circ + \sin 40^\circ) + \sin 40^\circ}{\cos 20^\circ}$$

$$\Rightarrow \frac{2\sin 30^\circ \cos 10^\circ + \sin 40^\circ}{\cos 20^\circ} \Rightarrow \frac{\cos 10^\circ + \cos 50^\circ}{\cos 20^\circ}$$

$$\Rightarrow \frac{2\cos 30^\circ \cos 20^\circ}{\cos 20^\circ} = \sqrt{3}$$

Summation of Trigonometric Series

To find sum of sines and cosines whose angles are in arithmetic progression.

(i) $\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \cdots + \sin\{\alpha + (n-1)\beta\}$

$$= \frac{\sin(n\beta/2)}{\sin(\beta/2)} \sin\left\{\alpha + \frac{n-1}{2}\beta\right\}$$

(ii) $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots + \cos\{\alpha + (n-1)\beta\}$

$$= \frac{\sin(n\beta/2)}{\sin(\beta/2)} \cos\left\{\alpha + \frac{n-1}{2}\beta\right\}$$

Q1 **Evaluate: sin1° + sin3° + sin5° + ... + sin89°**

Sol. Here $\alpha = 1^\circ$, $\beta = 2^\circ$

n = number of terms $= \frac{89+1}{2} = 45$

Now by summation formula of sine

$$\text{Given sum} = \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \sin\left(\alpha + \frac{n-1}{2}\beta\right)$$

$$= \frac{\sin(45^\circ)}{\sin(1^\circ)} \times \sin(45^\circ) = \frac{1/2}{\sin 1^\circ} = \frac{1}{2}\operatorname{cosec} 1^\circ$$

Q2 **Find the value of $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$**

Sol. Here $\alpha=\frac{2\pi}{7},\beta=\frac{4\pi}{7}-\frac{2\pi}{7}=\frac{2\pi}{7}$

n = number of terms = 3

$$\text{Given sum}=\frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}\cos\left(\alpha+\frac{n-1}{2}\beta\right)=\frac{\sin\left(\frac{3\pi}{7}\right)}{\sin\left(\frac{\pi}{7}\right)}\cos\left(\frac{4\pi}{7}\right)$$

$$=\frac{\sin(3\pi/7)\cos(\pi-3\pi/7)}{\sin(\pi/7)}$$

$$=\frac{-2\sin(3\pi/7)\cos(3\pi/7)}{2\sin\pi/7}=-\frac{\sin 6\pi/7}{\sin\pi/7}$$

$$=-\frac{\sin\left(\pi-\frac{\pi}{7}\right)}{\sin\left(\frac{\pi}{7}\right)}=-1$$

Q3 **The average of $2\sin^{\circ}2, 4\sin4^{\circ}, 6\sin6^{\circ}, \ldots, 180\sin180^{\circ}$ is**
(A) $\tan1^{\circ}$ **(B) $\cot1^{\circ}$** **(C) $\cos1^{\circ}$** **(D) $\sin1^{\circ}$**

Sol. **(B)**

$$\text{Average}=\frac{2\sin2^{\circ}+4\sin4^{\circ}+6\sin6^{\circ}+\cdots+180\sin180^{\circ}}{90}$$

$$=\frac{1}{90}\{(2\sin2^{\circ}+178\sin178^{\circ})+(4\sin4^{\circ}+176\sin176^{\circ})$$

$$+\ldots+(88\sin88^{\circ}+92\sin92^{\circ})+90\sin90^{\circ}+180\sin180^{\circ}\}$$

$$=\frac{1}{90}\{(180\sin2^{\circ})+(180\sin4^{\circ})+\ldots+(180\sin88^{\circ})+90+0\}$$

$$=2(\sin2^{\circ}+\sin4^{\circ}+\cdots+\sin88^{\circ})+1$$

$\because n=44,\alpha=2^{\circ},\beta=2^{\circ}$

$$=2\left\{\frac{\sin(44^{\circ})}{\sin1^{\circ}}\sin(45^{\circ})\right\}+1$$

$$= \left\{\frac{\cos1^\circ - \cos89^\circ}{\sin1^\circ}\right\} + 1$$

$$= \cot1^\circ - \frac{\sin1^\circ}{\sin1^\circ} + 1 = \cot1^\circ$$

Q4 **The value of**

$$(\cos^4 1^\circ + \cos^4 2^\circ + \cos^4 3^\circ + \ldots + \cos^4 179^\circ) - \left(\sin^4 1^\circ + \sin^4 2^\circ + \sin^4 3^\circ + \cdots + \sin^4 179^\circ\right)$$

(A) $2\cos1^\circ$ **(B) -1** **(C) $2\sin1^\circ$** **(D) 0**

Sol. **(B)**

$\cos^4\theta - \sin^4\theta = \left(\cos^2\theta + \sin^2\theta\right)\left(\cos^2\theta - \sin^2\theta\right) = 1 \times \cos2\theta$

Hence given expression

$\left(\cos^4 1^\circ - \sin^4 1^\circ\right) + \left(\cos^4 2^\circ - \sin^4 2^\circ\right) + \cdots + \left(\cos^4 179^\circ - \sin^4 179^\circ\right)$

$\Rightarrow \cos2^\circ + \cos4^\circ + \cos6^\circ + \cdots + \cos358^\circ$

$\Rightarrow 2\left(\cos2^\circ + \cos4^\circ + \cos6^\circ + \cdots + \cos178^\circ\right) + \cos180^\circ$

$n = 89,\ \alpha = 2^\circ,\ \beta = 2^\circ$

$$\Rightarrow 2\frac{\sin89^\circ}{\sin1^\circ}\cos90^\circ + (-1) = -1$$

Q5 **Find n if sum $\sin\frac{\pi}{n} + \sin\frac{2\pi}{n} + \sin\frac{3\pi}{n} + \ldots + \sin(n-1)\frac{\pi}{n} = 2 + \sqrt{3}$**

Sol. On LHS

$$\alpha = \frac{\pi}{n},\ \beta = \frac{2\pi}{n} - \frac{\pi}{n} = \frac{\pi}{n}$$

Number of terms $m = (n - 1)$

$$\therefore \quad \text{LHS} = \frac{\sin\left(\frac{m\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}\sin\left(\alpha + \frac{n-1}{2}\beta\right)$$

$$= \frac{\sin\left\{\frac{(n-1)\pi}{2n}\right\}}{\sin\left\{\frac{\pi}{2n}\right\}}\sin\left(\frac{\frac{n\pi}{n}}{2}\right)$$

$$= \frac{\sin\left(\frac{\pi}{2} - \frac{\pi}{2n}\right)}{\sin \pi / 2n} = \frac{\cos \frac{\pi}{2n}}{\sin \frac{\pi}{2n}} = \cot \frac{\pi}{2n}$$

Now $\cot \frac{\pi}{2n} = 2 + \sqrt{3} = \cot 15° = \cot \frac{\pi}{12}$

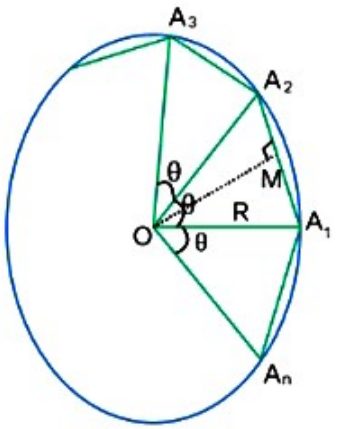
$\Rightarrow 2n = 12 \Rightarrow n = 6$

Q6 **In a regular polygon of n-sides with $A_1, A_2, A_3 \ldots\ldots A_n$ vertices prove that**

$$\left(A_1A_2\right)^2 + \left(A_1A_3\right)^2 + \cdots + \left(A_1A_n\right)^2 = 2nR^2$$

where R is radius of circumcircle circumscribing it.

Sol. $\theta = \frac{2\pi}{n} \Rightarrow n\theta = 2\pi$ ...(i)



In Δ OMA$_1$

$MA_1 = R \sin \frac{\theta}{2}$ $\quad \therefore A_1A_2 = 2R \sin \frac{\theta}{2}$

$$\left(A_1A_2\right)^2 = 2R^2\left(2\sin^2 \frac{\theta}{2}\right)$$

$$= 2R^2\left(1 - \cos\theta\right)$$

Similarly, $\left(A_1A_3\right)^2 = 2R^2\left(1 - \cos 2\theta\right)$

$$\left(A_1A_n\right)^2 = 2R^2\left(1 - \cos(n-1)\theta\right)$$

$$\left(A_1A_2\right)^2 + \left(A_1A_3\right)^2 + \cdots + \left(A_1A_n\right)^2$$

$$= 2R^2\{(n-1) - \cos\theta - \cos 2\theta - \cdots - \cos(n-1)\theta]$$

$$= 2R^2\left\{(n-1) - \frac{\sin(n-1)\frac{\theta}{2}}{\sin\left(\frac{\theta}{2}\right)}\cos\left(\frac{n\theta}{2}\right)\right\}$$

$$= 2R^2\left\{(n-1) - \frac{\sin\left(\pi - \frac{\pi}{n}\right)\cos\pi}{\sin\left(\frac{\pi}{n}\right)}\right\}$$

$$= 2R^2\left\{(n-1) - (-1)\right\} = 2nR^2$$

Hence proved.

Summation of Miscellaneous Series

Splitting the sum series as difference of 2 terms.

Q1 **Prove that** $\tan\frac{x}{2}\sec x + \tan\frac{x}{2^2}\sec\frac{x}{2} + \tan\frac{x}{2^3}\sec\frac{x}{2^2} + \ldots$ **(upto n terms)**

$= \tan x - \tan\frac{x}{2^n}$

Sol. $\tan\frac{\theta}{2}\cdot\sec\theta = \tan\frac{\theta}{2}\cdot\frac{1}{\cos\theta}$

$$= \tan\frac{\theta}{2}\left\{\frac{1+\tan^2\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}\right\}$$

$$= \tan\frac{\theta}{2}\left[\frac{-1+\tan^2\frac{\theta}{2}+2}{1-\tan^2\frac{\theta}{2}}\right\}$$

$$= \tan\frac{\theta}{2}\left\{-1+\frac{2}{1-\tan^2\frac{\theta}{2}}\right\}$$

$$= -\tan\frac{\theta}{2} + \frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}} \Rightarrow \tan\theta - \tan\frac{\theta}{2}$$

$\therefore \tan\frac{x}{2}\sec x = \tan x - \tan\frac{x}{2}$ ...(i)

$\tan\frac{x}{2^2}\sec\frac{x}{2} = \tan\frac{x}{2} - \tan\frac{x}{2^2}$ ...(ii)

$\vdots \qquad \vdots \qquad \vdots$

$\tan\frac{x}{2^n}\sec\frac{x}{2^{n-1}} = \tan\frac{x}{2^{n-1}} - \tan\frac{x}{2^n}$ ...(n)

On adding all equations we get

$$\tan\frac{x}{2}\sec x + \tan\frac{x}{2^2}\sec\frac{x}{2} + \tan\frac{x}{2^3}\sec\frac{x}{2} + \ldots + \tan\frac{x}{2^n}\sec\frac{x}{2^{n-1}} = \tan x - \tan\frac{x}{2^n}$$

Hence proved.

Q2 **Prove that sinx sec3x + sin 3x sec9x + sin9x sec27x ... upto n-terms**

$= \frac{1}{2}\left\{\tan 3^n x - \tan x\right\}$

Sol. $\sin\theta \sec 3\theta = \frac{\sin\theta}{\cos 3\theta} = \frac{2\sin\theta\cos\theta}{2\cos\theta.\cos 3\theta}$

$\Rightarrow \frac{\sin(3\theta - \theta)}{2\cos\theta\cos 3\theta}$

$\Rightarrow \frac{\sin 3\theta\cos\theta - \cos 3\theta\sin\theta}{2\cos\theta\cos 3\theta}$

$\Rightarrow \frac{1}{2}\{\tan 3\theta - \tan\theta\}$

$\therefore \quad \sin x \sec 3x = \frac{1}{2}\left\{\tan 3x - \tan x\right\}$ ...(i)

$\sin 3x \sec 3^2 x = \frac{1}{2}\left\{\tan 3^2 x - \tan 3x\right\}$ ...(ii)

$\vdots \qquad \vdots \qquad \vdots$

$\sin 3^{n-1} x \sec 3^n x = \frac{1}{2}\left\{\tan 3^n x - \tan 3^{n-1} x\right\}$ ...(n)

On adding all equation we get

$\sin x \sec 3x + \sin 3x \sec 9x + \sin x \sec 27x + \ldots\ldots + \sin 3^{n-1} x \sec 3^n x$

$= \frac{1}{2}\left\{\tan 3^n x - \tan x\right\}$ **Hence proved.**

Q3 **Prove that:**

$$\textbf{cosec } x + \textbf{cosec} 2x + \textbf{cosec} 4x + \ldots + \textbf{cosec} 2^n x = \cot\frac{x}{2} - \cot 2^n x$$

Sol. $\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{\sin\frac{\theta}{2}}{\sin\frac{\theta}{2}\sin\theta}$

$\Rightarrow \frac{\sin\left(\theta - \frac{\theta}{2}\right)}{\sin\frac{\theta}{2}.\sin\theta} = \frac{\sin\theta\cos\frac{\theta}{2} - \cos\theta\sin\frac{\theta}{2}}{\sin\frac{\theta}{2}\sin\theta}$

$\Rightarrow \cot\frac{\theta}{2} - \cot\theta$

$\therefore$ $\operatorname{cosec} x = \cot\frac{x}{2} - \cot x$ ...(i)

$\operatorname{cosec} 2x = \cot x - \cot 2x$ ...(ii)

$\vdots \quad \vdots \quad \vdots$

$\operatorname{cosec} 2^n x = \cot 2^{n-1}x - \cot 2^n x$ ...(n)

On adding all equations we get

$$\operatorname{cosec} x + \operatorname{cosec} 2x + \operatorname{cosec} 4x + \ldots + \operatorname{cosec} 2^n x = \cot\frac{x}{2} - \cot 2^n x$$

Hence proved.

Inequalities in a triangle:

Q1 **In any ΔABC show that $\cot^2 A + \cot^2 B + \cot^2 C \geq 1$**

Sol. $(\cot A - \cot B)^2 + (\cot B - \cot C)^2 + (\cot C - \cot A)^2 \geq 0$

$(\cot^2 A + \cot^2 B + \cot^2 C) \geq 1$

Hence proved.

Q2 **In any ΔABC show that $\Sigma \sin A \leq \frac{3\sqrt{3}}{2}$**

Sol. By using graph of sinx from $x \in (0, \pi)$

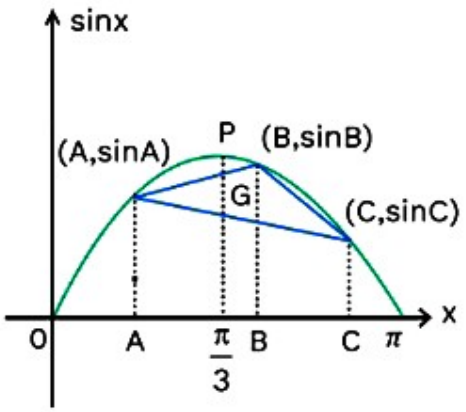
Centroid of the triangle ABC is

$G\left(\frac{A+B+C}{3}, \frac{\sin A + \sin B + \sin C}{3}\right)$

$\left(\frac{\pi}{3}, \frac{\sin A + \sin B + \sin C}{3}\right)$

Now value of sinx at is $x = \frac{\pi}{3}$ is $\sin\frac{\pi}{3}$

$\therefore P(\pi/3, \sqrt{3}/2)$



Clearly from graph

$$\frac{\sin A + \sin B + \sin C}{3} \leq \frac{\sqrt{3}}{2}$$

Q3 **In Δ ABC prove that $\cos A \cos B \cos C \leq \frac{1}{8}$**

Sol. Let $\cos A \cos B \cos C = \frac{1}{2}\{\cos(A+B)+\cos(A-B)\}\cos C$

$$= \frac{1}{2}\{\cos(\pi - C)+\cos(A-B)\}\cos C$$

$$= \frac{1}{2}\{-\cos^2 C + \cos(A-B)\cos C\} = x$$

$\Rightarrow \quad \cos^2 C - \cos(A-B)\cos C + 2x = 0$

Which is quadratic in cosC

$\therefore \quad D \geq 0$

$$\{\cos(A-B)\}^2 - 4(1)(2x) \geq 0$$

$$8x \leq \cos^2(A-B) \leq 1$$

$$8x \leq 1 \quad x \leq \frac{1}{8}$$

Hence proved.

Q4 **In ΔABC prove that $1 < \cos A + \cos B + \cos C \leq 3/2$**

Sol. Let $\cos A + \cos B + \cos C = x$

$$2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2} = x$$

$$2\sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2} = x$$

$$\Rightarrow \quad 2\sin^2\frac{C}{2} - 2\cos\left(\frac{A-B}{2}\right)\sin\frac{C}{2} + (x-1) = 0 \qquad \ldots(i)$$

Which is quadratic in $\sin\frac{C}{2}$ hence

$$D \geq O$$

$$4\cos^2\left(\frac{A-B}{2}\right) - 4(2)(x-1) \geq 0$$

$$\cos^2\left(\frac{A-B}{2}\right) - 2(x-1) \geq 0$$

$$2(x-1) \leq \cos^2\left(\frac{A-B}{2}\right) \leq 1$$

$\Rightarrow \quad x \le \frac{3}{2}$

Also product of roots of equation (i) will be positive

$\therefore \quad \left(\frac{x-1}{2}\right) > 0 \quad \Rightarrow x > 1$

Hence $1 < x \le \frac{3}{2}$ **Hence proved.**

Elimination

To obtain parametric representation of curves we use elimination for e.g., to obtain trajectory of a moving object we have to eliminate the parameter 't' from both equation given in x and y to get a relation in x and y.

This relation give us the equation of curve

Parametric co- ordinates:

(i) For $x^2 + y^2 = r^2 \Rightarrow x = r\cos\theta,\ y = r\sin\theta$ ($\theta \rightarrow$ parameter)

(ii) For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow x = a\cos\theta, y = b\sin\theta$ ($\theta \rightarrow$ parameter)

(iii) For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow x = a\sec\theta,\ y = b\tan\theta$ ($\theta \rightarrow$ parameter)

Q1 **Suppose a particle is moving on a circle with center (0,0) and radius 2. Then at any point it will be at a position with co-ordinate (2cosθ, 2sinθ). Find the equation of curve which the particle travels.**

Sol. According to give information

$x = 2\cos\theta$ ...(i)

$y = 2\sin\theta$ ...(ii)

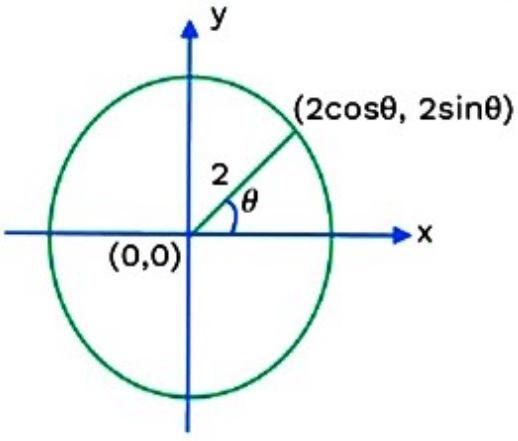
Now to eliminate θ

$(i)^2 + (ii)^2$

$x^2 + y^2 = 2^2\left(\cos^2\theta\right) + 2^2\left(\sin^2\theta\right)$

$x^2 + y^2 = 2^2$

Which is equation of required curve.



Q2 **Find the maximum and minimum value of $3x + 4y$ if $x^2 + y^2 = 1$**

Sol. $\because\ x^2 + y^2 = 1$

Let $x = \cos\theta, \quad y = \sin\theta$

Now $3x + 4y = 3\cos\theta + 4\sin\theta$

$$-\sqrt{3^2 + 4^2} \le 3\cos\theta + 4\sin\theta \le \sqrt{3^2 + 4^2}$$

$\therefore$ minimum of $3x + 4y = -5$

maximum of $3x + 4y = 5$

Q3 **Find the maximum and minimum value of $3x + 4y$ if $\frac{x^2}{4} + \frac{y^2}{9} = 1$.**

Sol. $\because\ \frac{x^2}{2^2} + \frac{y^2}{3^2} = 1$

$\therefore$ Let $x = 2\cos\theta, \quad y = 3\sin\theta$

Now $3x + 4y = 6\cos\theta + 12\sin\theta$

$$-\sqrt{6^2 + 12^2} \le 6\cos\theta + 12\sin\theta \le \sqrt{6^2 + 12^2}$$

$\therefore\ (3x + 4y)_{min} = -6\sqrt{5}$

$(3x + 4y)_{max} = 6\sqrt{5}$

Q4 **Find the maximum and minimum value of $3x - y$ if $\frac{(x-1)^2}{3} + \frac{(y-2)^2}{4} = 4$**

Sol. $\because\ \frac{(x-1)^2}{12} + \frac{(y-2)^2}{16} = 1 \Rightarrow \frac{(x-1)^2}{\left(2\sqrt{3}\right)^2} + \frac{(y-2)^2}{4^2} = 1$

$\therefore$ let $x - 1 = 2\sqrt{3}\cos\theta,\ y - 2 = 4\sin\theta$

$x = 1 + 2\sqrt{3}\cos\theta,\ y = 2 + 4\sin\theta$

Now $3x - y = 3\left(1 + 2\sqrt{3}\cos\theta\right) - (2 + 4\sin\theta)$

$= 1 + 6\sqrt{3}\cos\theta - 4\sin\theta$

$$(3x - y)_{min} = 1 - \sqrt{\left(6\sqrt{3}\right)^2 + 4^2} = 1 - 2\sqrt{31}$$

$$(3x - y)_{max} = 1 + \sqrt{\left(6\sqrt{3}\right)^2 + 4^2} = 1 + 2\sqrt{31}$$

Q5 **Eliminate θ between the given equations $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = c^2$ and $l\tan\theta = m$**

Sol. $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = c^2$ ...(i)

$\frac{l}{\cos\theta} - \frac{m}{\sin\theta} = 0$ ...(ii)

(i) × l – (ii) × ax

$$\frac{amx}{\sin\theta} - \frac{bly}{\sin\theta} = lc^2$$

$$\Rightarrow \quad \sin\theta = \left(\frac{amx - bly}{lc^2}\right) \qquad ...(iii)$$

Now from (ii) $\cos\theta = \frac{l}{m}\sin\theta$

$\because \quad \sin^2\theta + \cos^2\theta = 1$

$$\sin^2\theta + \frac{l^2}{m^2}\sin^2\theta = 1 \Rightarrow \sin^2\theta\cdot\left(1 + \frac{l^2}{m^2}\right) = 1$$

from (iii) $$\frac{(amx - bly)^2}{l^2c^4}\cdot\left(\frac{l^2 + m^2}{m^2}\right) = 1$$

Q6 **If $\cos\theta > 0$, $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$ then show that $m^2 - n^2 = 4\sqrt{mn}$**

Sol. $m^2 - n^2 = (\tan\theta + \sin\theta)^2 - (\tan\theta - \sin\theta)^2$

$= 4\tan\theta\sin\theta$ ...(i)

$mn = (\tan\theta + \sin\theta)(\tan\theta - \sin\theta)$

$= \tan^2\theta - \sin^2\theta$

$$= \frac{\sin^2\theta}{\cos^2\theta} - \sin^2\theta = \sin^2\theta\left(\sec^2\theta - 1\right)$$

$= \sin^2\theta\cdot\tan^2\theta$

$4\sqrt{mn} = 4\sin\theta\cdot\tan\theta \qquad (\because \cos\theta > 0)$ ...(ii)

From (i) and (ii)

$m^2 - n^2 = 4\sqrt{mn}$ **Hence proved.**

Q7 **If $\operatorname{cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$, eliminate θ**

Sol. $\dfrac{1}{\sin\theta} - \sin\theta = m \Rightarrow \dfrac{1-\sin^2\theta}{\sin\theta} = m$

$\Rightarrow \quad \cos^2\theta = m\sin\theta$ ...(i)

$\dfrac{1}{\cos\theta} - \cos\theta = n \Rightarrow \dfrac{1-\cos^2\theta}{\cos\theta} = n$

$\Rightarrow \quad \sin^2\theta = n\cos\theta$ ...(ii)

$m^2n = \dfrac{\cos^4\theta}{\sin^2\theta} \cdot \dfrac{\sin^2\theta}{\cos\theta} = \cos^3\theta$

$n^2m = \dfrac{\sin^4\theta}{\cos^2\theta} \cdot \dfrac{\cos^2\theta}{\sin\theta} = \sin^3\theta$

$\therefore \quad \left(m^2n\right)^{2/3} + \left(n^2m\right)^{2/3} = 1$